Guided Tutorial for Pentaho Data Integration using IBM DB2

In the data integration exercise, you will use the Pentaho Data Integration tool to transform two data sources and load data into a DB2 fact table. You will perform transformations to parse date strings, combine fields, and perform validation checks. Before starting this tutorial, you need to install necessary software, check/prepare data sources, and create tables used in the tutorial.

1. Tutorial Prerequisites

Before starting this tutorial, you should download and install the server and client for IBM DB2 Cloud. Refer to IBM DB2 Installation Instructions

You also need to install Pentaho Data Integration before starting this tutorial. After installing Pentaho Data Integration, you need to install (or connect) the Java Database Connectivity (JDBC) driver for DB2. Refer to Pentaho Data Integration and DB2 JDBC Driver Installation . This tutorial demonstrates the community edition of the most recent stable version (5.0.1) of Pentaho Data Integration.

After installing Pentaho Data Integration, you need to obtain the data sources:

- Excel file used (**SSExcelSource.xls**) in part 1 of the tutorial
- Access database used (**SSAccessData.mdb**) in part 2 of the tutorial

The tutorial uses the Store Sales data warehouse as depicted in Figure 1. Sales is the entity type surrounded by 1-M relationships with dimension entity types, Item, Customer, Store, and TimeDim. The schema design has a snowflake for the 1-M relationship from Division to Store. In the table design, table names have been preceded with the prefix “SS” to avoid conflicts with other tables. Thus, the fact table is SSSales, not Sales as shown in the ERD of Figure 1.

You need to run batch files to create the database. Use this steps to populate the data warehouse:

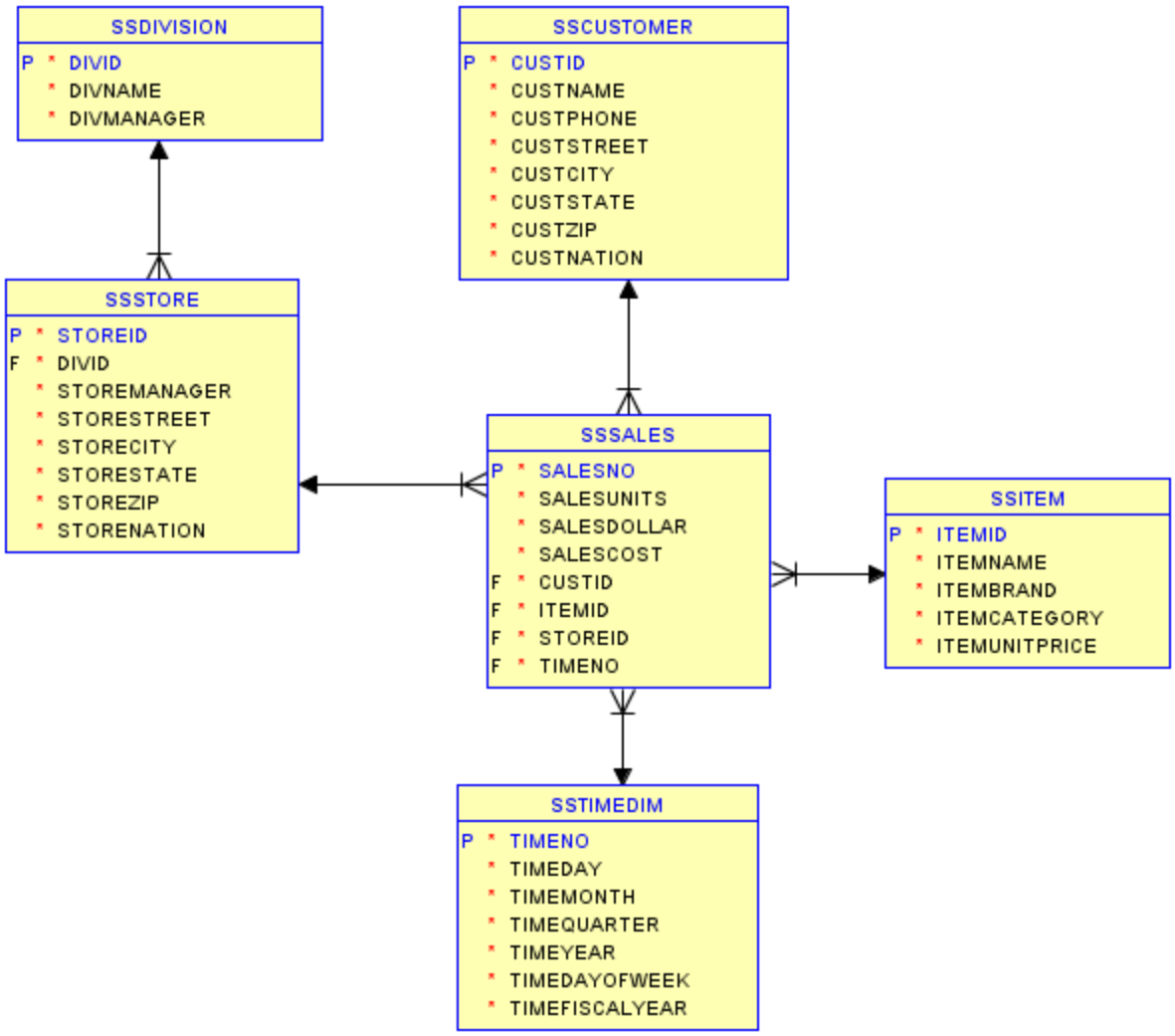


Figure 1: DB2 Snowflake Schema for the Store Sales Data Warehouse

2. Creating your First Transformation

The Data Integration component of Spoon allows you to create transformations and jobs. Transformations involve data flows such as reading from a source, transforming data and loading it into a target location. Jobs coordinate transformations such as defining dependencies among transformations and execution conditions such as, “Is my source file available?” or “Does a table exist in my database?

This exercise will step you through building your first transformation with Pentaho Data Integration introducing common concepts along the way. Follow the instructions below to create a new transformation.

1. After starting Pentaho Data Integration, you will see the opening window (Figure 2) and the Spoon window (Figure 3).

2. Click  (New) in the upper left corner of the Spoon window.

3. Select **Transformation** from the list of components (Figure 4) displayed after selecting the **New** button.

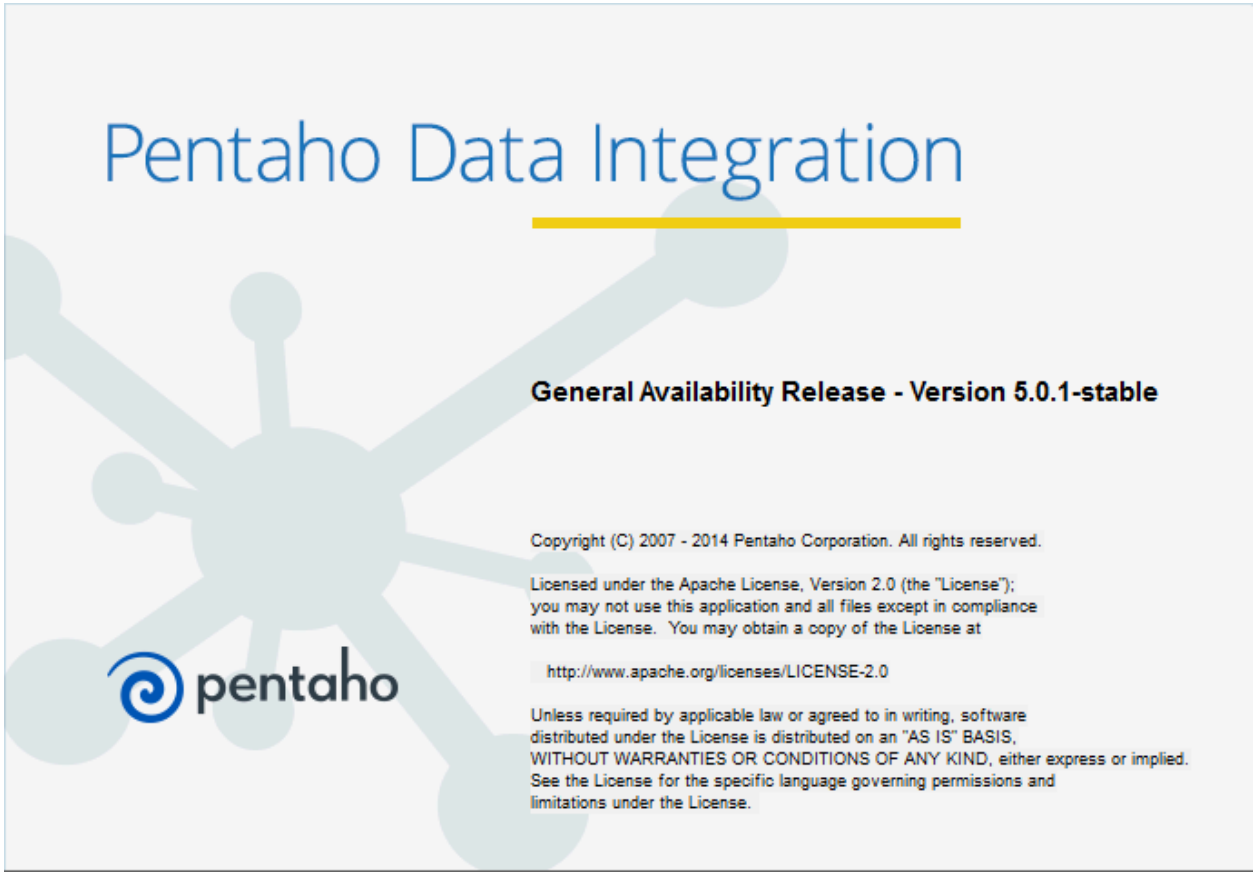


Figure 2: Pentaho Data Integration Welcome Window

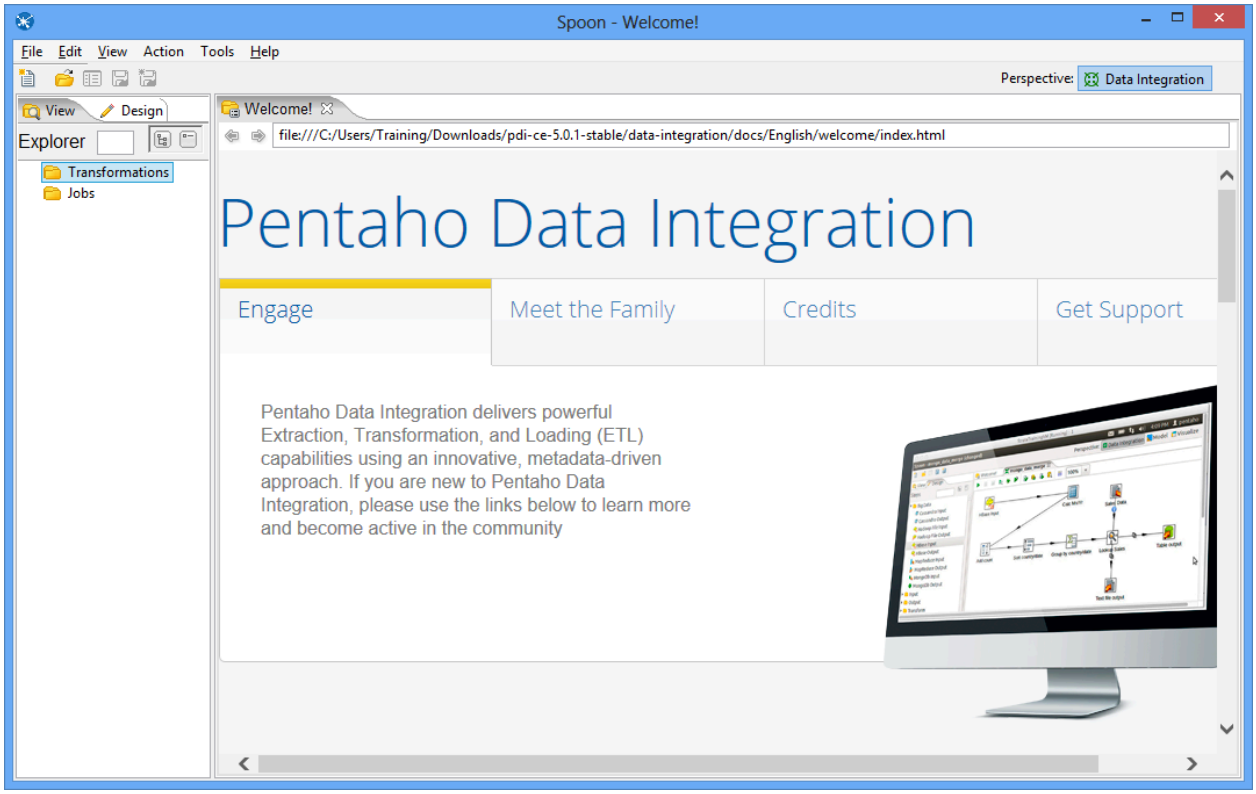


Figure 3: Spoon Opening Window

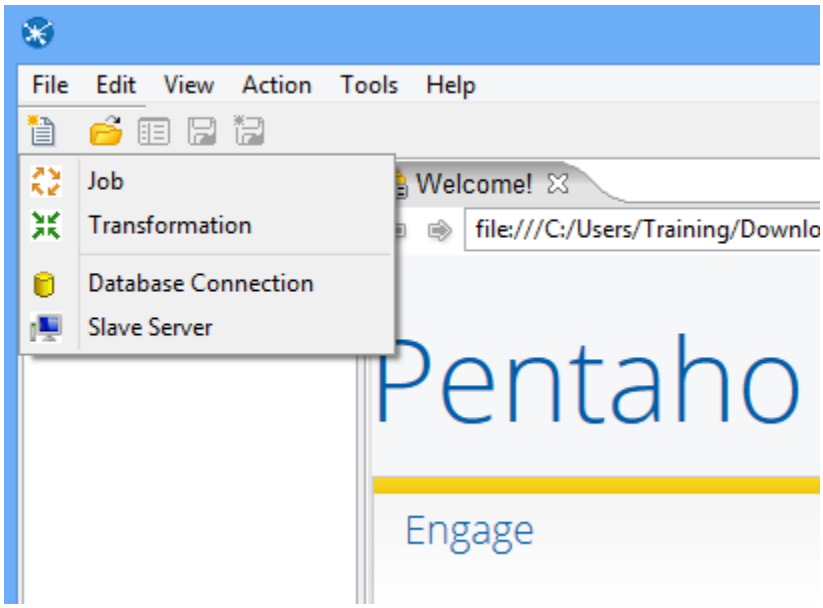


Figure 4: Spoon Transformation List

PART 1: Excel File as Data Source

3. Load the first data source from Excel

Make sure that you have SSExcelSource.xls the Excel input file. You need to know the location of this file in Step 4 below.

Step 1 – In the View tab, right click the new transformation 1 and select "settings…"

Step 2 – Set the Transformation name for the new transformation as: **SSExcelSource_[surname]** and click OK.

Step 3 – Save the transformation following **File** → **Save.** Save the file under your *C:\..\Documents.* You will see the empty transformation window in the Spoon (Figure 5).

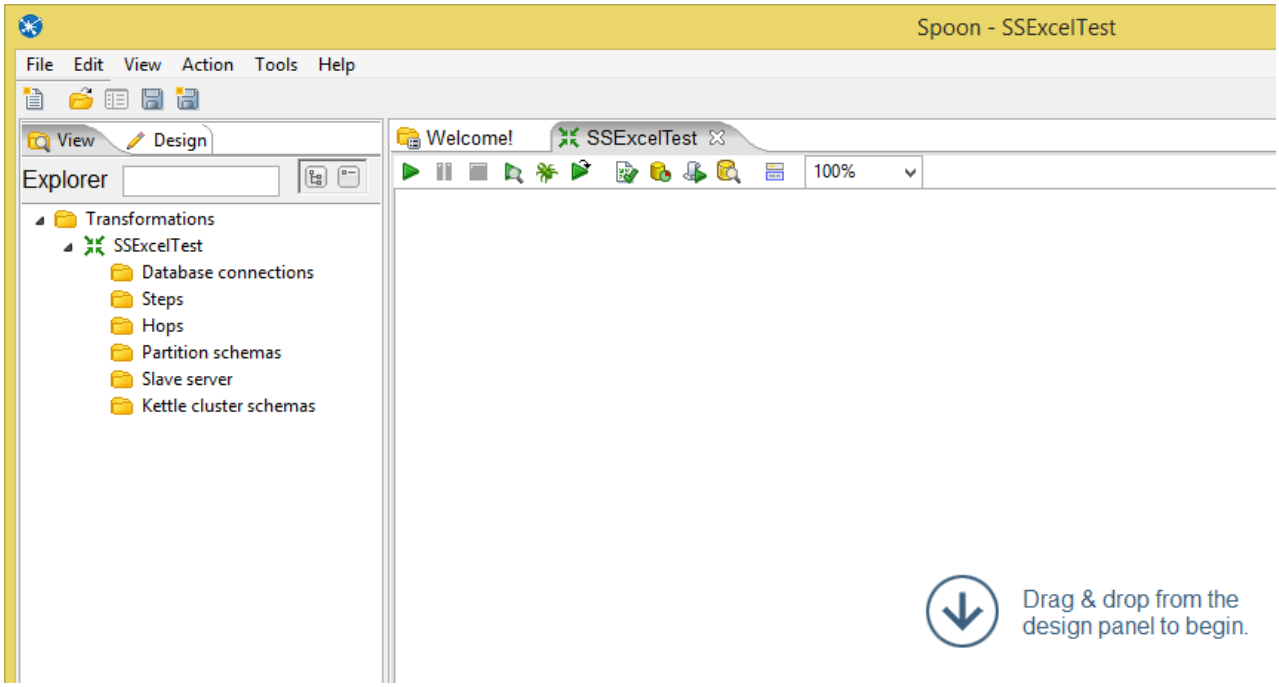


Figure 5: Empty Transformation Window

Step 4 – Create the Excel Input step:

Step 4.1 Under the **Design** tab, expand the Input node (Figure 6).

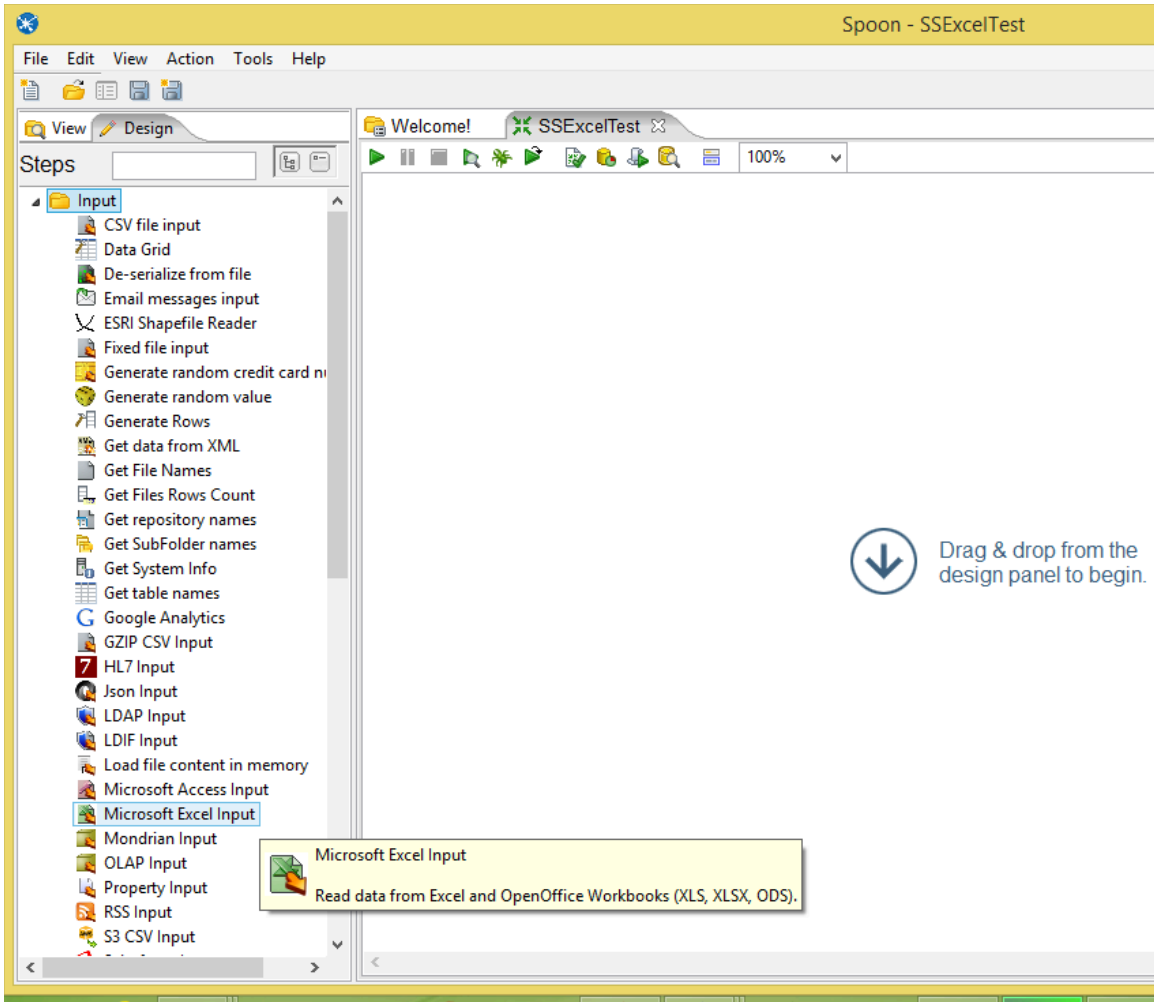


Figure 6: New Microsoft Excel Input Node

Step 4.2 Select and drag a **Microsoft Excel Input** step into the canvas on the right.

Step 4.3 Double Click on the **Microsoft Excel Input** step. The edit properties dialog box (Figure 7) associated with the **Microsoft Excel Input** step appears.

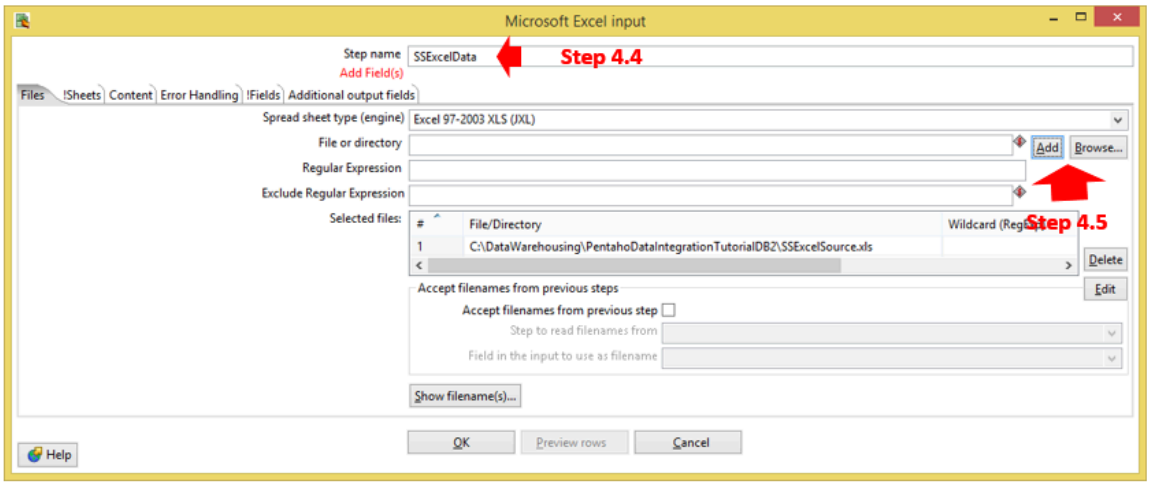


Figure 7: Files Window for Microsoft Excel Input Property Editing

Step 4.4 Set name for the Excel Input as **SSExcelData**. Download the SSExcelSource.xls

Step 4.5 Click the button "***Browse***…" and locate the Excel **SSExcelSource.xls**. Then, Click "***Add***" to add the file to the selected files area.

Step 4.6 In the tab named **Sheets**, click the button "**Get sheetname(s)…**". There will appear an **Enter List** (Figure 8) to choose sheets. Select **Sheet 1**, press "**>**" to move it into the right area. Click **OK**.

Step 4.7 In the tab names **Fields,** click on **"Get fields from header row…"** You need to change the data types, length, and precision as the specification in Figure 9.

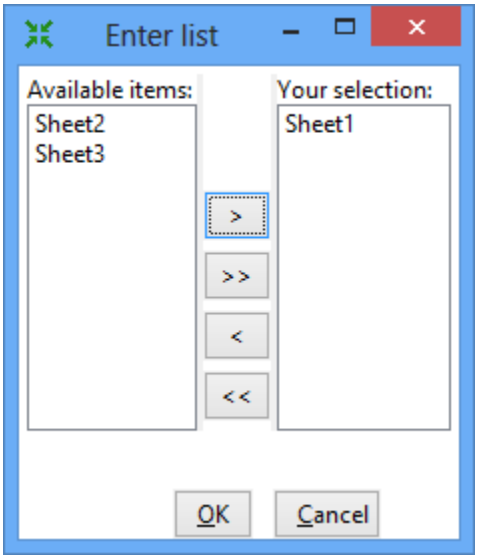


Figure 8: Sheet Specification Window

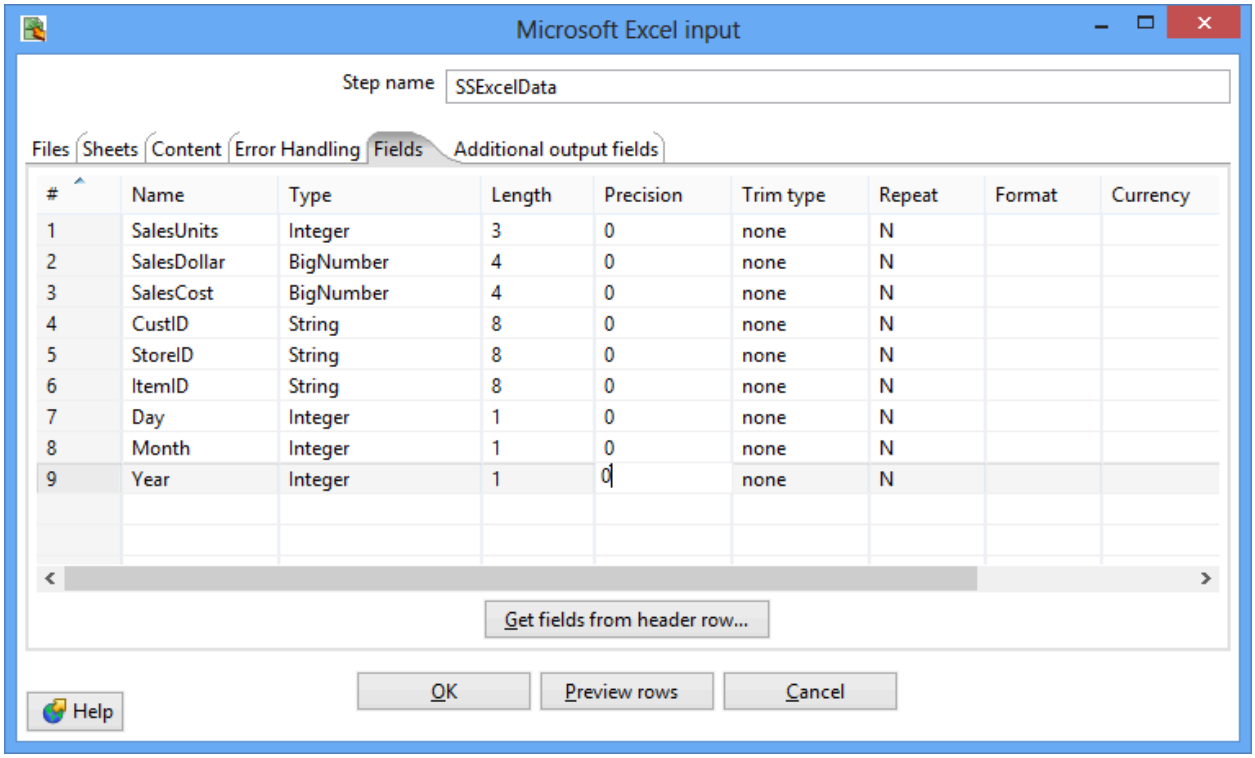


Figure 9: Fields Window for Microsoft Excel Input Property Editing

Step 4.8 Click **OK** at the bottom of the window. The input icon will change to the **SSExcelData** icon displayed in Figure 10.

Step 4.9 To check the output of what you have accomplished, click **SSExcelData** icon.

Run the **Preview this transformation** icon  as you see in Figure 11. Click **Quick launch** button as **Transformation debug dialog** opens. Figure 12 shows the output of the transformation. This is the actual content of Excel spreadsheet **SSExcelSource.xls** then named **SSExcelData** transformation name. Close the two succeeding windows when done checking the preview data

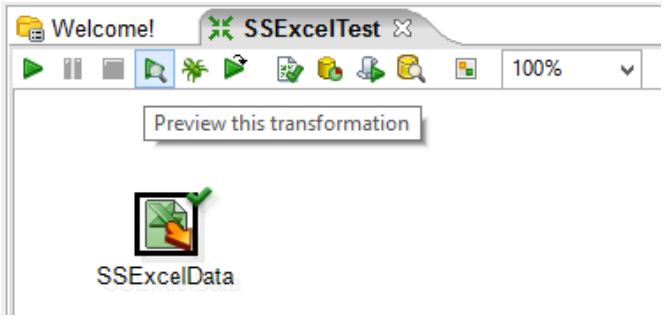
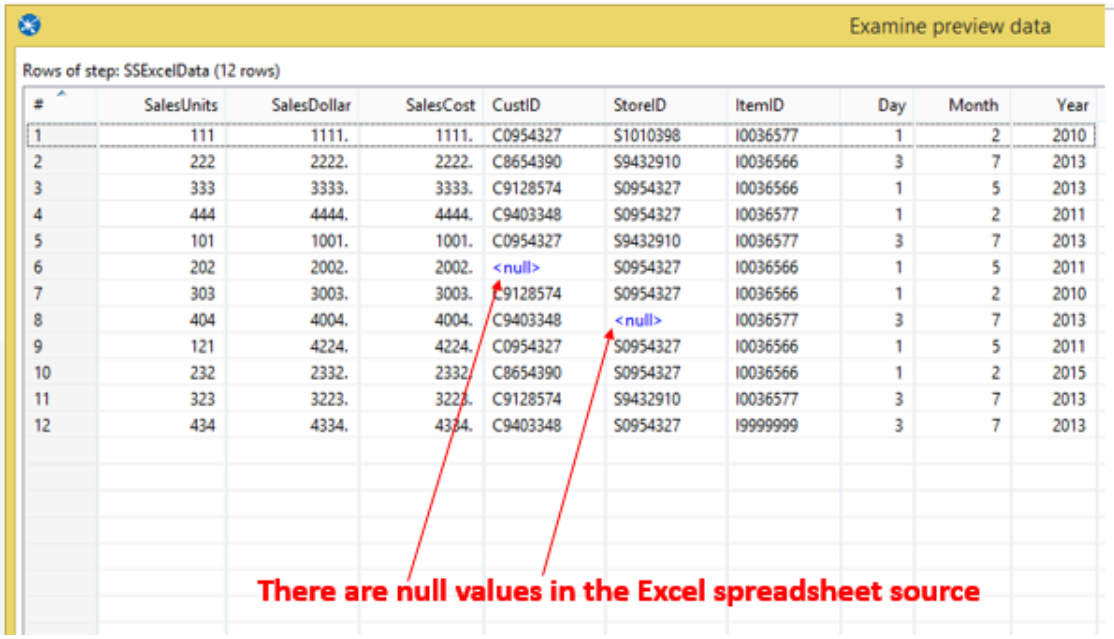


Figure 11. Selecting the Excel source and check its output transformation



#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year
1	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010
2	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013
3	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013
4	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011
5	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013
6	202	2002.	2002.	<null>	S0954327	I0036566	1	5	2011
7	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010
8	404	4004.	4004.	C9403348	<null>	I0036577	3	7	2013
9	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011
10	232	2332.	2332.	C8654390	S0954327	I0036566	1	2	2015
11	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013
12	434	4334.	4334.	C9403348	S0954327	I9999999	3	7	2013

Figure 12. Content of SSExcelSource.xls

Step 5 – In this part of the tutorial, you will add constraint checking for null values and appropriate data types for the Excel data source.

Step 5.1 Add a Filter Rows step to your transformation. Under the **Design** tab, go to **Flow** → **Filter Rows** and drag it in the canvas (Figure 13).

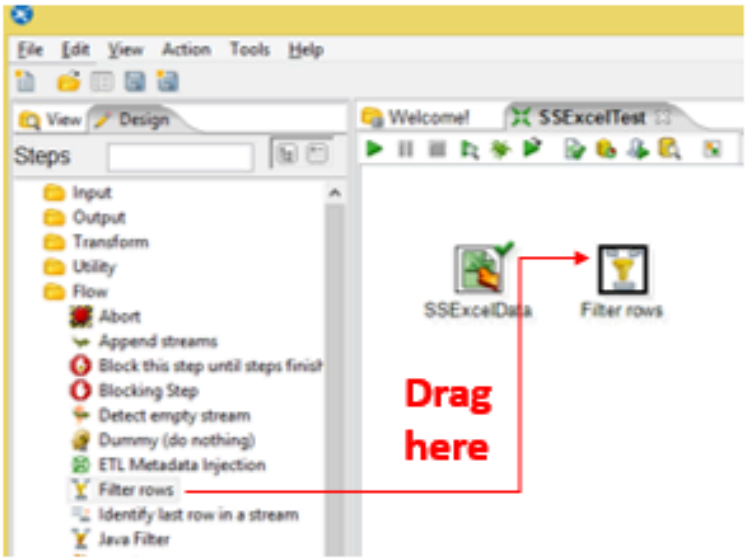


Figure 13: Excel Input Node and Filter Node in Spoon

Step 5.2 Create a "hop" between the **SSExcelData** (Excel SSExcelSource file input) step and the **Filter Rows** step. Hops are used to describe the flow of data in your transformation. To create the hop, click the **SSExcelData** step, draw a line from right arrow icon towards the **Filter Rows** step (Figure 14).

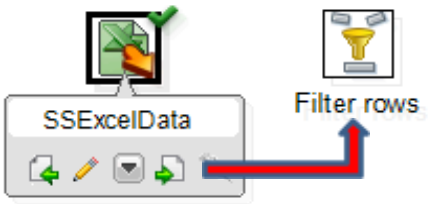


Figure 14: Hop connecting an Excel Input Node Connected to a Filter Node

Step 5.3 Double-click the **Filter Rows** step. The **Filter Rows** edit properties dialog box appears (Figure 15).

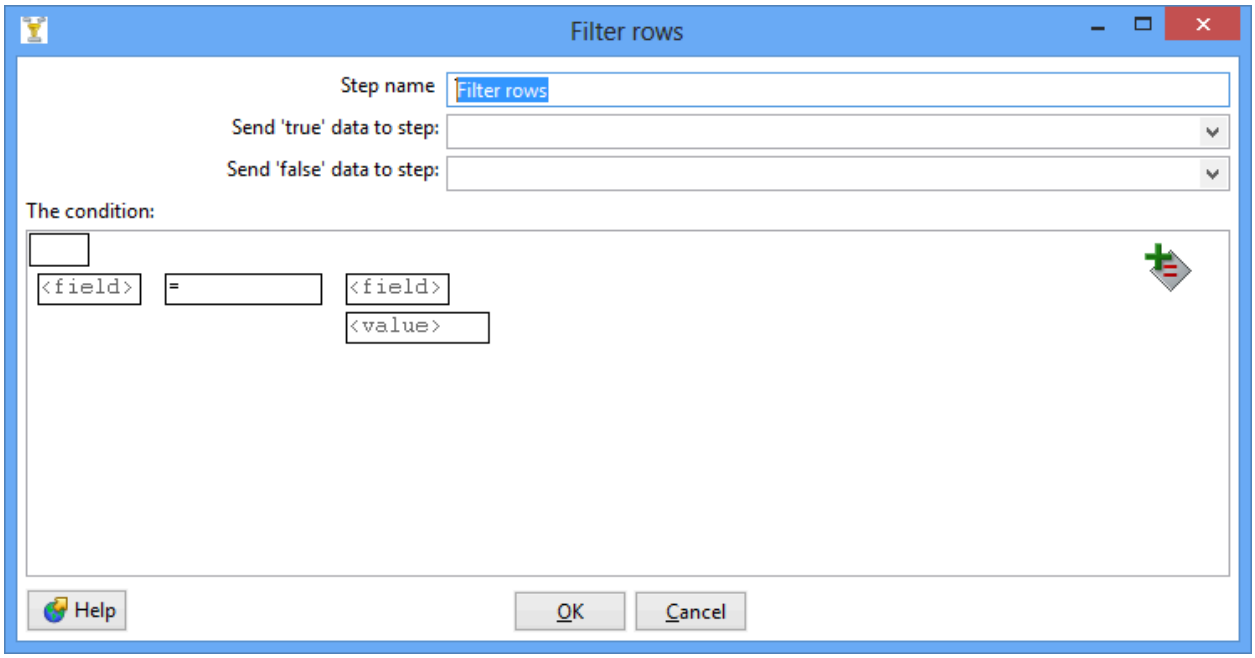


Figure 15: Property Edit Window of Filter Node

Step 5.4 Rename **Step Name** field with **Filter rows by default**.

Step 5.5 Under **The condition**, click <field>. A dialog box that contains the fields you can use to create your condition appears.

Step 5.6 In the **Fields**: dialog box (Figure 16) select **SalesUnits** and click **OK**.

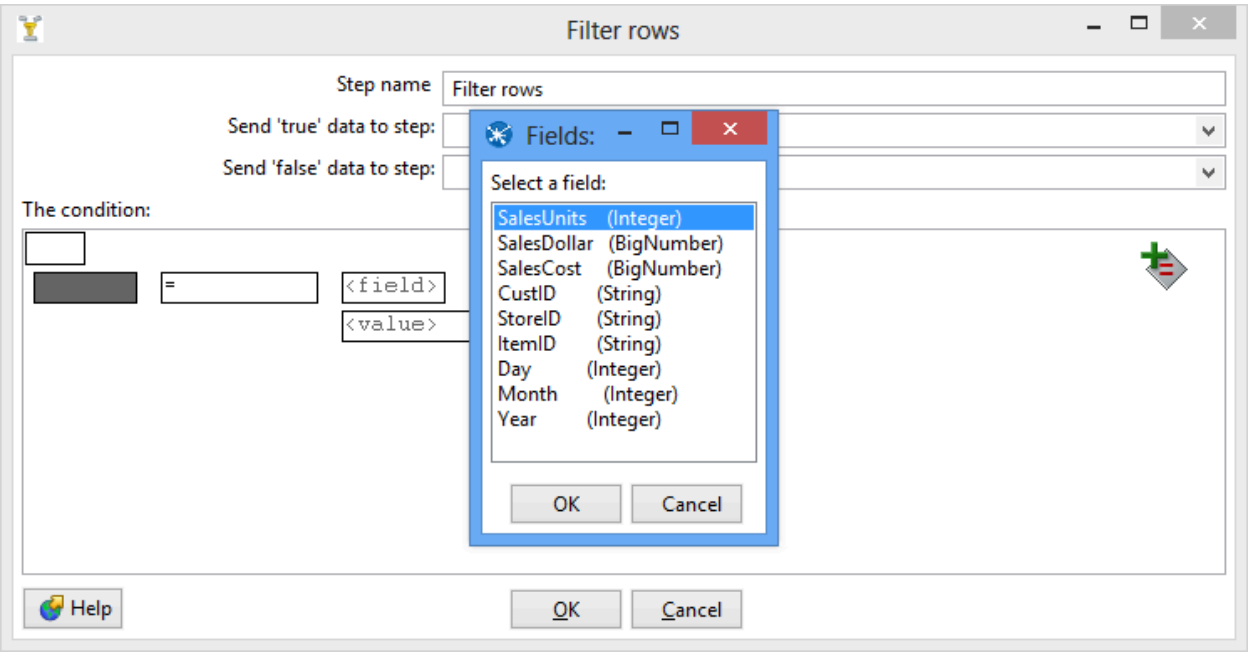


Figure 16: Condition Fields Selection Window

Step 5.7 Click on the comparison operator (Figure 17) (set to = by default) and select the **IS NOT NULL** function and click **OK**.

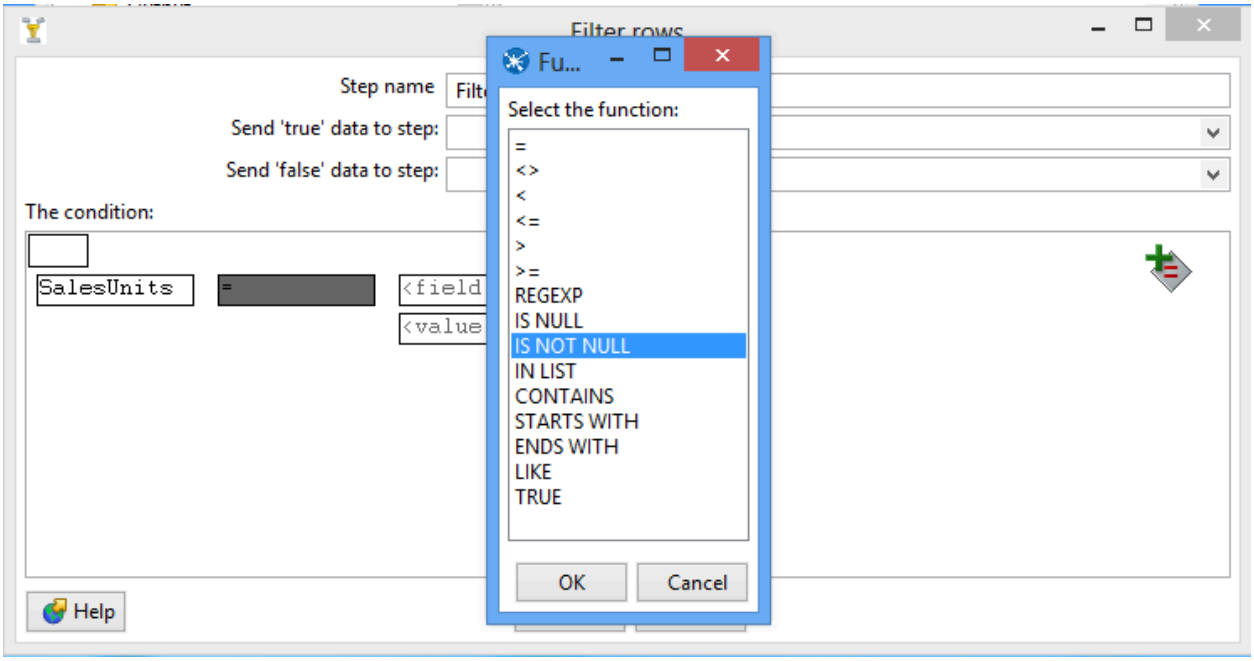


Figure 17: Comparison Operator List

Step 5.8 Click the button  . A new condition row appears with **null = [ ]** as a default.

Step 5.9 Click on the expression and add constraints for the next column similarly to what you did for "**SalesUnits**"

Step 5.10 Click on **UP**. This will allow you to see both conditions joint by AND

Step 5.11 Click the button  again. Another new condition row appears with **null = [ ]** as a default.

Step 5.12 Keeping repeating these steps for all fields.

Step 5.13 The final view of filter conditions is shown by Figure 18. Click OK button to close window

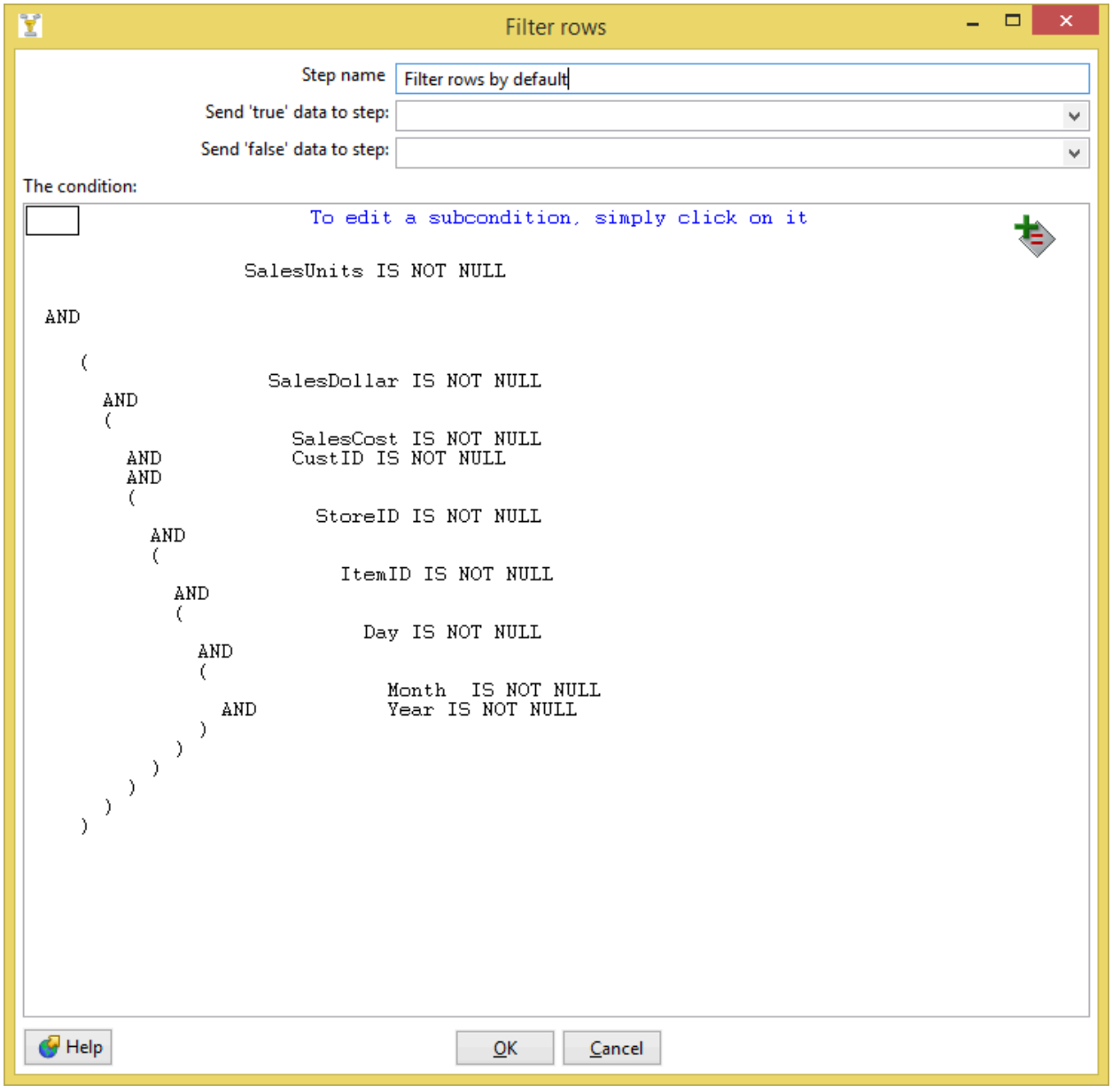


Figure 18: Filter Conditions Window

Step 5.14 Click **Run this transformation** icon again. Click **Quick Launch** button to view the output of the transformation. Figure 19 shows the output of **Filter rows by default** removing rows with NULL values that were forwarded by **SSExcelData**.

Examine preview data

Rows of step: Filter rows by default (10 rows)

#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year
1	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010
2	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013
3	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013
4	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011
5	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013
6	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010
7	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011
8	232	2332.	2332.	C8654390	S0954327	I0036566	1	2	2015
9	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013
10	434	4334.	4334.	C9403348	S0954327	I9999999	3	7	2013

Figure 19. NULL values were removed from the Filter Rows by default transformation

Step 5.15 Close preview data by clicking the two succeeding **Close** buttons.
Step 5.16 Save the current file.

Step 6 – Create a step to sort the result of the Filter Rows step.

Step 6.1 Under the **Design** tab, expand the contents of the **Transform** node.
Step 6.2 Click and drag a **Sort Rows** step into your transformation; create a hop between the **Filter rows** and Sort Rows steps. Select **Result is TRUE** in the filter results selection list (Figure 20).

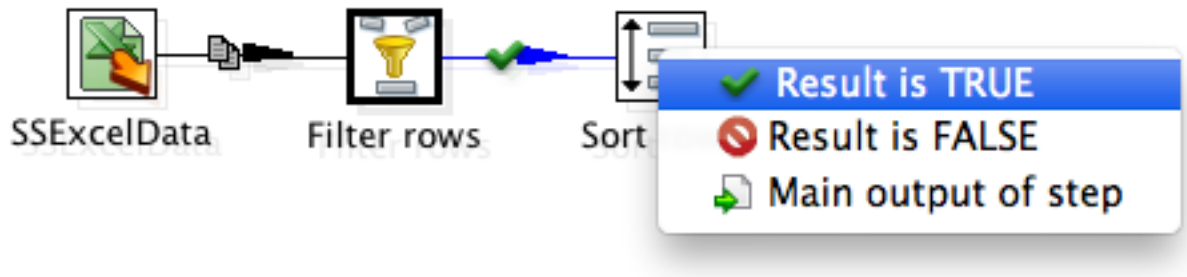


Figure 20: Filter Results Selection List

Step 6.3 Double-click the **Sort Rows** step to open its edit properties dialog box (Figure 21). Click "**Get Fields**" to obtain the fields. Delete other fields except the Day, Month and Year fields. To delete fields, you can multiple select by clicking the row number (selecting the entire row) holding the SHIFT KEY (or CTRL key) then click the next row number, right click any of those selected rows, and click **Delete selected lines**. Then click Ok.

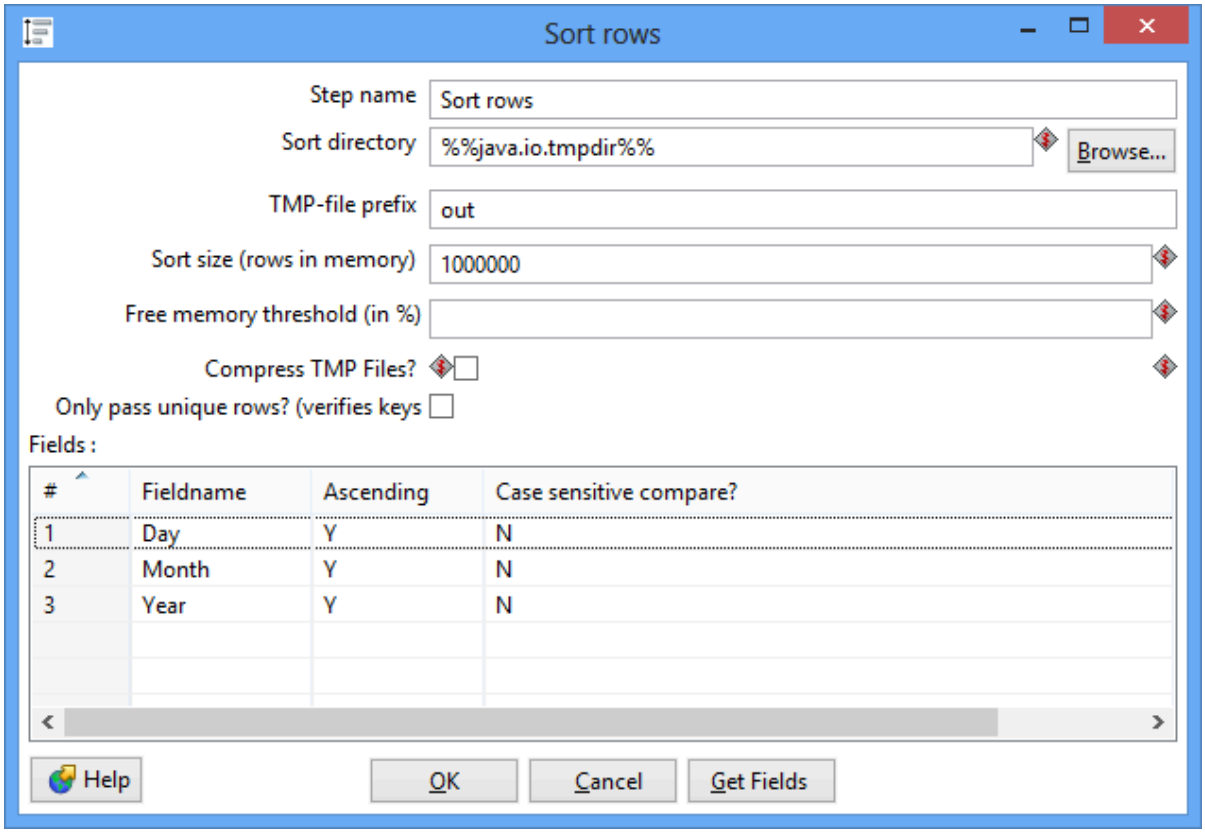
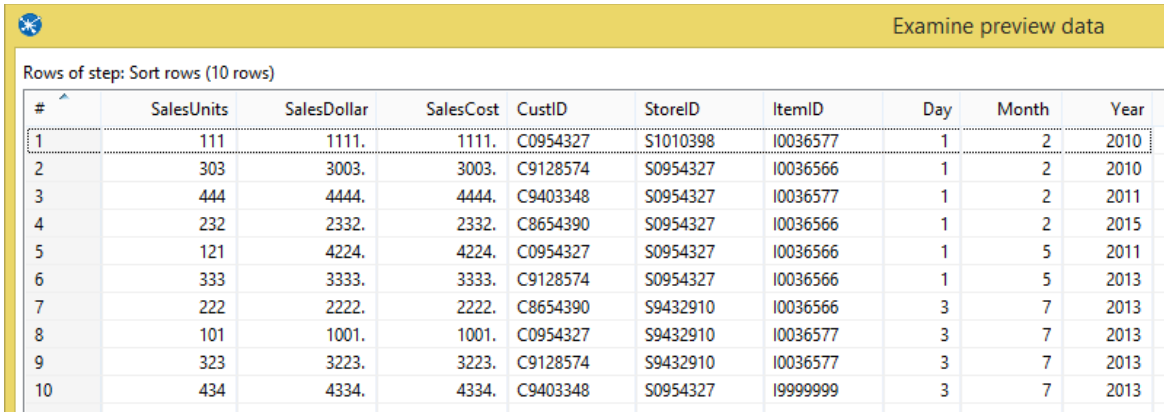


Figure 21: Property Edit Window of Sort Rows Node

Step 6.4 Click **Run this transformation** icon  again. Click **Quick Launch** button to view the output of the transformation. Figure 22 shows the output of **Sort rows** sorting rows by DAY, MONTH, YEAR. Please check the output.



#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year
1	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010
2	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010
3	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011
4	232	2332.	2332.	C8654390	S0954327	I0036566	1	2	2015
5	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011
6	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013
7	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013
8	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013
9	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013
10	434	4334.	4334.	C9403348	S0954327	I9999999	3	7	2013

Figure 22: Rows are sorted by Day, Month, Year

Step 6.5 Close preview data by clicking the two succeeding **Close** buttons.
Step 6.6 Save file.

4. Using a Database Connection to Lookup Columns from DB2 tables

Pentaho Data Integration allows you to define connections to multiple databases provided by multiple database vendors (DB2, MySQL, Oracle, Postgres, and many more). Pentaho Data Integration ships with the most suitable JDBC drivers for supported databases and its primary interface to databases is through JDBC. Vendors write a driver that matches the JDBC specification and Pentaho Data Integration uses the driver. Unless you require extensive debugging or have other needs, you won't ever need to write your own database driver.

When you define a database connection, the connection information (username, password, port number, and so on) is stored in the Pentaho Enterprise Repository and is available to other users when they connect to the repository. If you are not using the Pentaho Enterprise Repository, the database connection information is stored in the XML file associated with a transformation or job.

Connections that are available for use with a transformation or job are listed under Database **Connection** node in the explorer **View** in Spoon.

Let us define a database connection.

Step 1 -- Click **View** tab.

Step 2 – Double click Database Connections. Figure 23 dialog appears.

Step 3 – Set *Connection name* with **DB2Connection**; *hostname* with your one designated hostname as stated in the video example ; *Port number* is <depending in what is listed in your

credentials>. Step 4 – Make sure **General**, **IBM DB2** is selected under *Connection Type*; **Native (JDBC)** under *Access*.

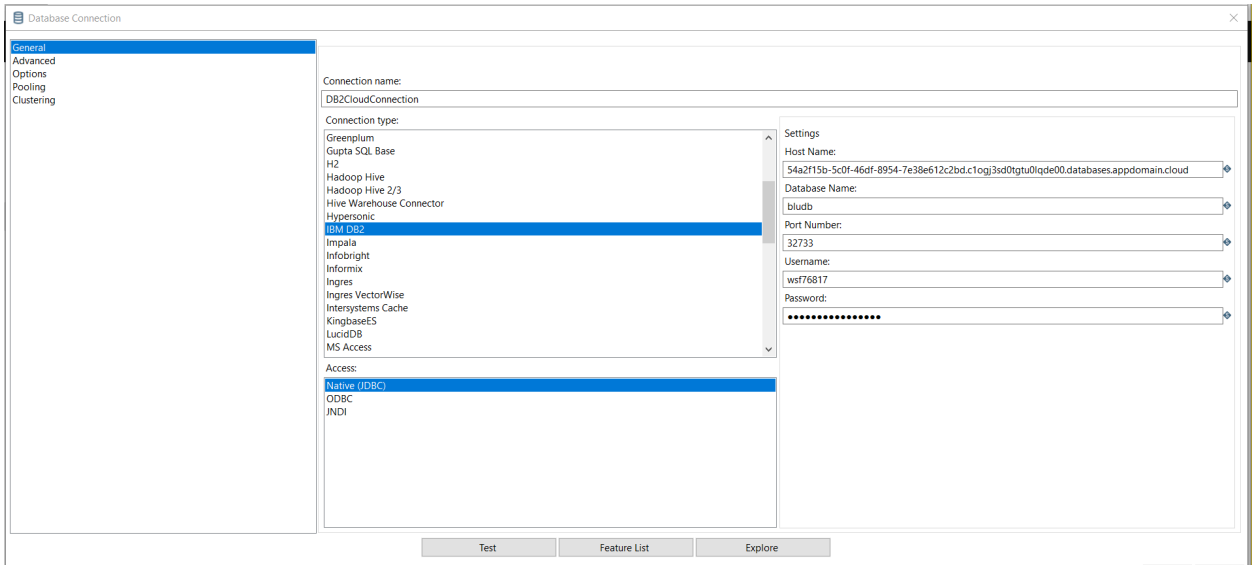


Figure 23: Setting up a database connection in DB2

Step 4.1 - Go to **Options** menu. Supply in the **Parameter** : sslConnection, **Value**: true.

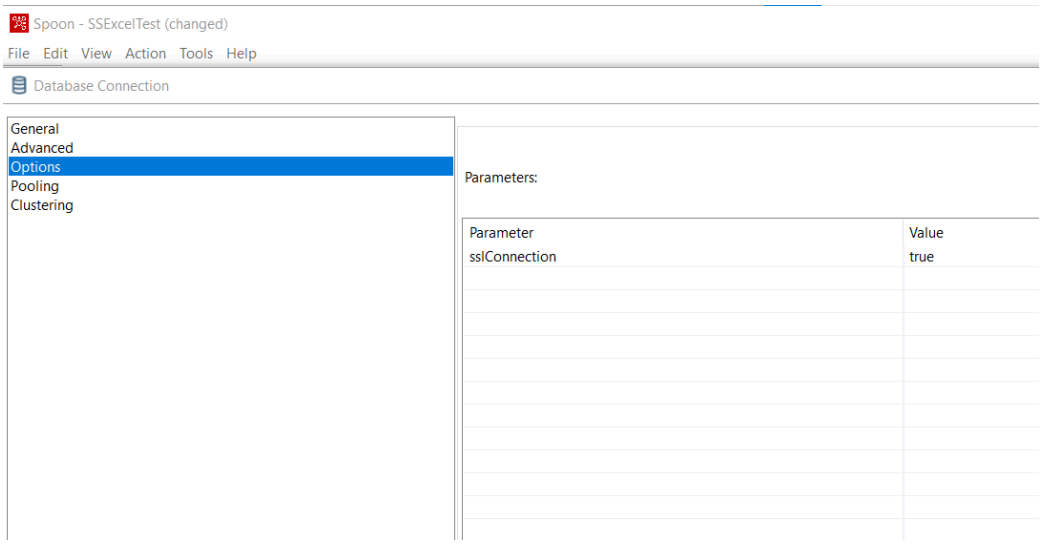


Figure 23.1: Setting the sslConnection to true.

Step 5 – Click Test button to check if our connection is successful. Figure 24 confirms the connection. Click succeeding **OK** buttons to close.

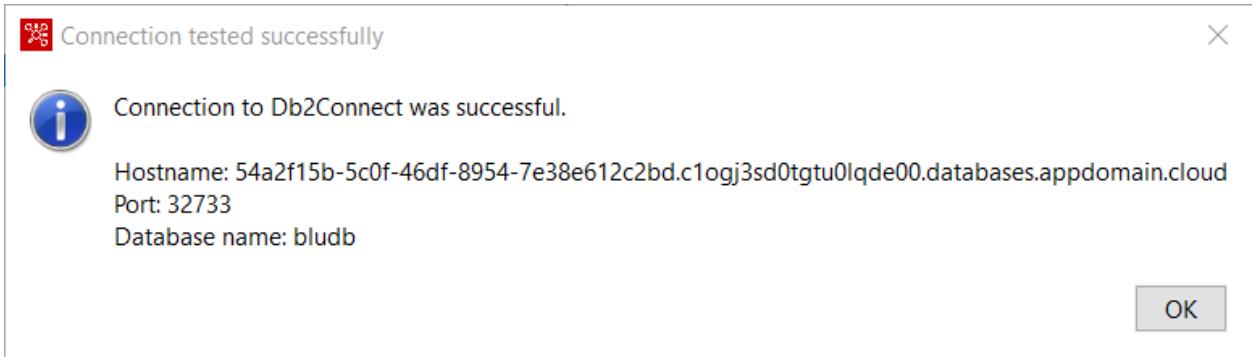


Figure 24: Successful database connection is established

This part of the tutorial involves looking up the date from the ***SSTimeDim*** table to check the validity of dates in the Excel data source. In addition, you will lookup primary key columns from other DB2 tables to ensure loaded data does not contain invalid foreign keys.

> **IMPORTANT NOTE:** Before setting the connection information, you should first configure the JDBC driver according to the instructions described in the installation procedure for Pentaho Data Integration. You must also have created and populated the tables of the Store Sales data warehouse.

Step 6 – Access the ***SSTimeDim*** table from DB2 database.

> Step 6. 1 Under the **Design** tab, expand the contents of the **Input** node.
> Step 6.2 Click and drag a **Table Input** step into your transformation.
> Step 6.3 Double-click the Table Input step to open its edit properties dialog box (Figure 25).
> Step 6.4 Rename your Table Input step name to *SSTimeDim*. You would notice **DB2Connection** is already supplied under *Connection name*. This is the effect of establishing the database connection first done at Step 3.
> Step 6.5 Type in "`SELECT * FROM SSTimeDim`" in the SQL section (Figure 25). SSTimeDim is the table name. You can click the **Preview** button to view the content of the table. Accept Preview size of 1000. Figure 26 displays the content of **SSTimeDim**. Click Ok, to exit the Database Connection dialog box.

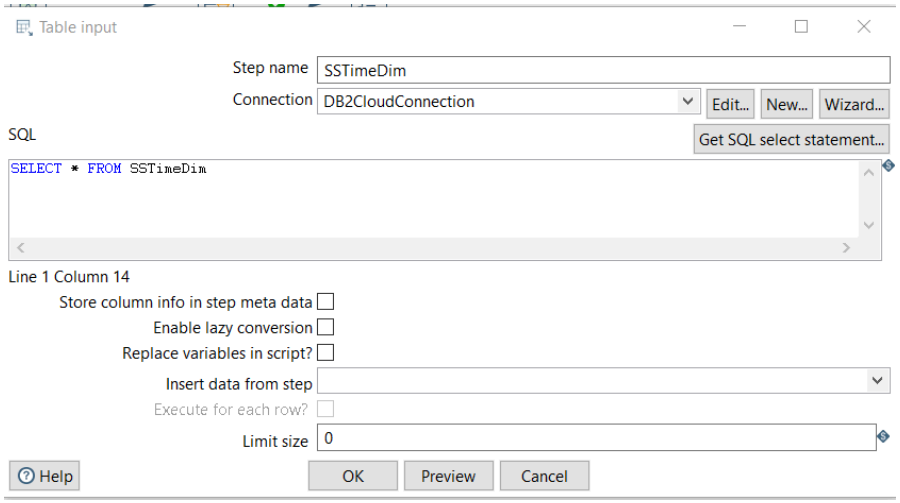
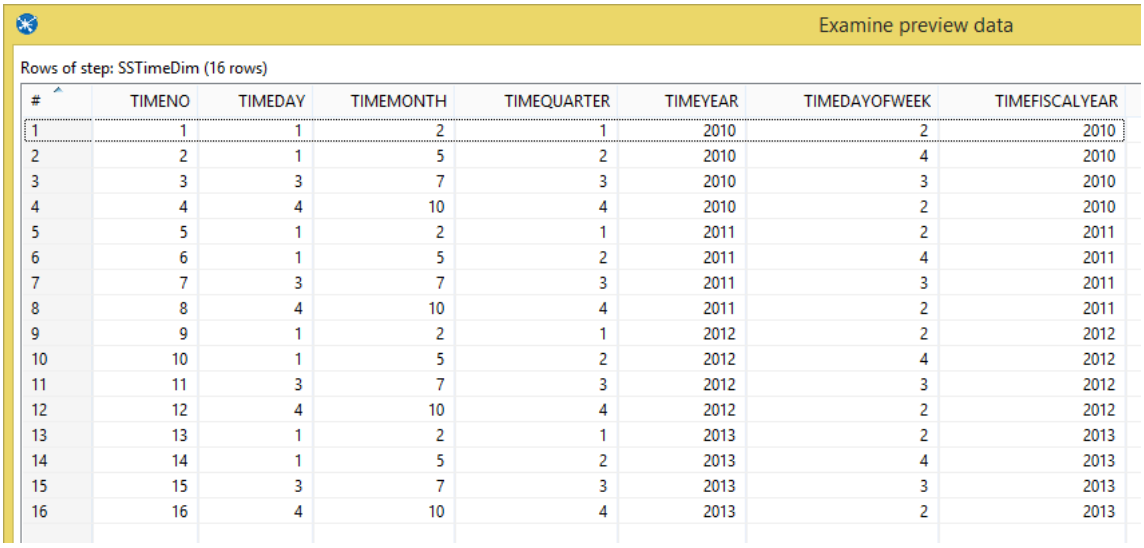


Figure 25: SQL Edit Section in Property Window of Table Input Node



#	TIMENO	TIMEDAY	TIMEMONTH	TIMEQUARTER	TIMEYEAR	TIMEDAYOFWEEK	TIMEFISCALYEAR
1	1	1	2	1	2010	2	2010
2	2	1	5	2	2010	4	2010
3	3	3	7	3	2010	3	2010
4	4	4	10	4	2010	2	2010
5	5	1	2	1	2011	2	2011
6	6	1	5	2	2011	4	2011
7	7	3	7	3	2011	3	2011
8	8	4	10	4	2011	2	2011
9	9	1	2	1	2012	2	2012
10	10	1	5	2	2012	4	2012
11	11	3	7	3	2012	3	2012
12	12	4	10	4	2012	2	2012
13	13	1	2	1	2013	2	2013
14	14	1	5	2	2013	4	2013
15	15	3	7	3	2013	3	2013
16	16	4	10	4	2013	2	2013

Figure 26: Content of SSTimeDim table

Step 6.6 Add another sort rows component **Sort rows 2**, and a hop connecting the *SSTimeDim* step. In the field specification (Figure 27), delete other fields except TIMEDAY, TIMEMONTH, TIMEYEAR fields.

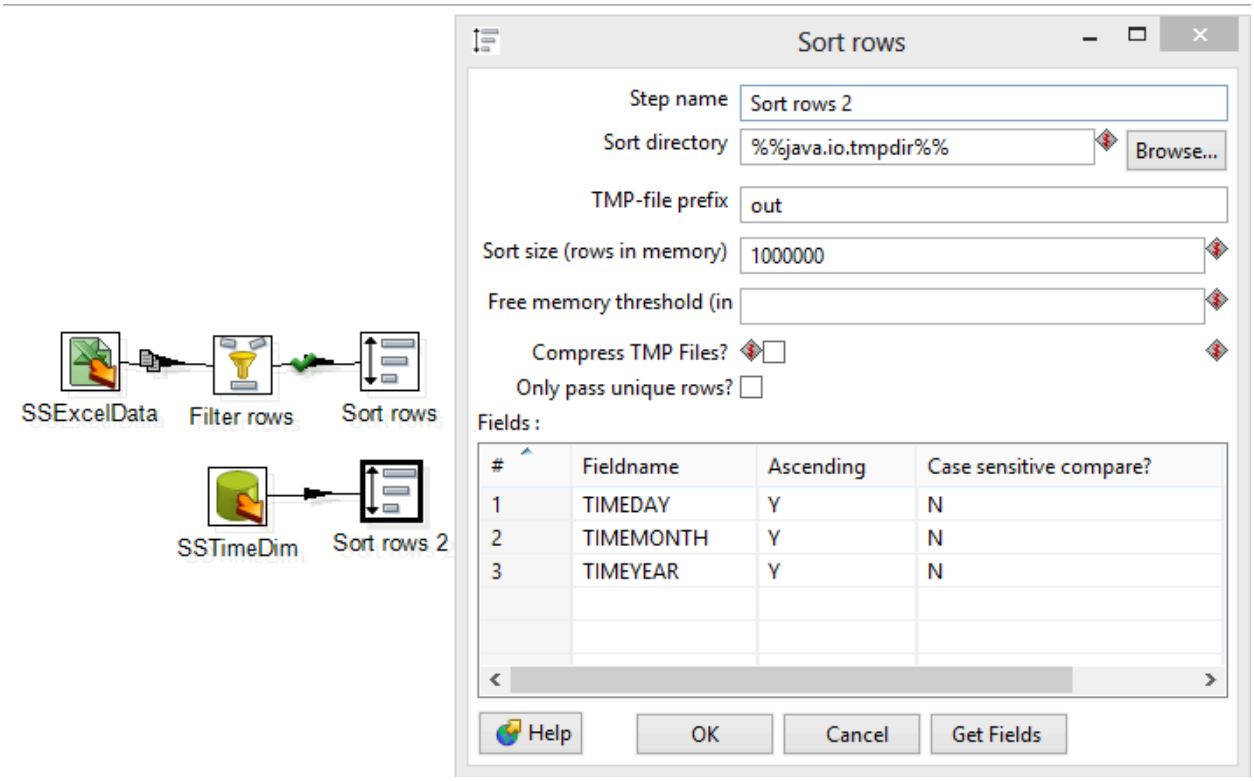


Figure 27: Property Edit Window of Sort Rows 2 Node

Step 6.7 Click **Run this transformation** icon again. Click the **Quick Launch** button to view the output of the transformation. Figure 28 shows the output of **Sort rows 2** sorting rows by TIMEDAY, TIMEMONTH, TIMEYEAR of *SSTimeDim*. Click **Close** buttons to exit preview data.

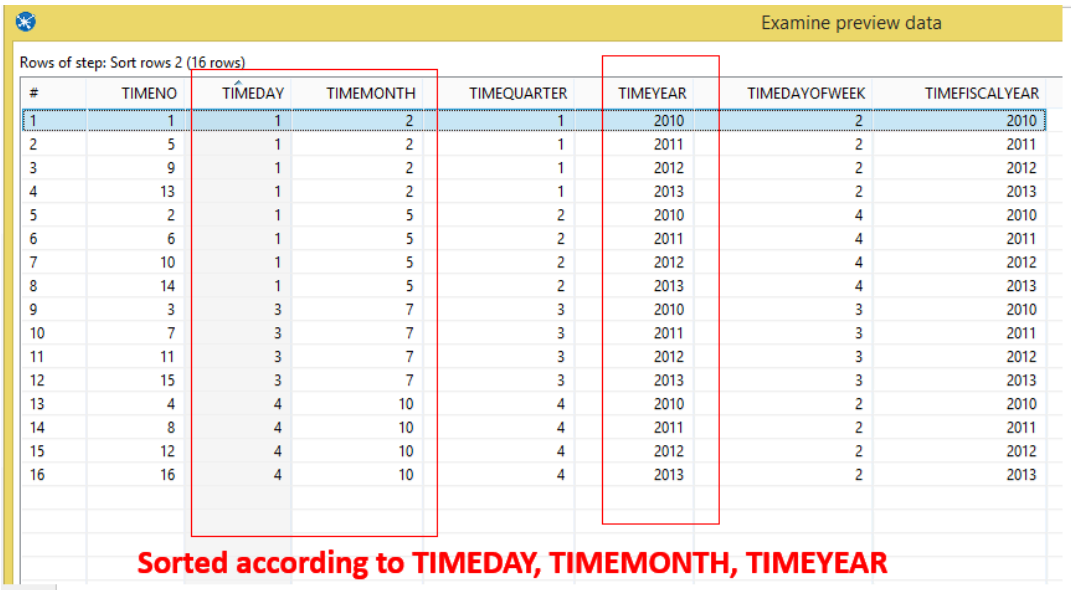


Figure 28: Sort rows 2 output

Step 6.7 Under the **Design** tab, expand the contents of the **Joins** node.

Step 6.8 Click and drag a **Merge Join** step into your transformation; create a hop between the **Sort rows, Sort rows 2** and **Merge Join** steps (Figure 29).

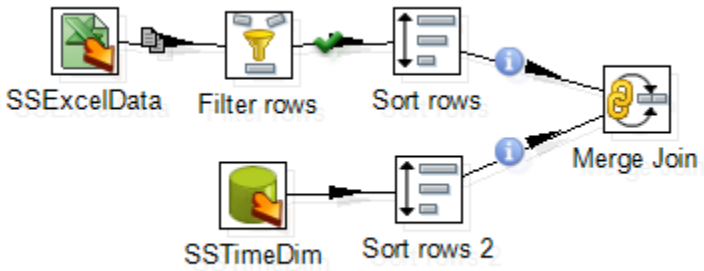


Figure 29: Two Sort Rows Nodes Connected to Merge Join Node

Step 6.9 Double-click the **Merge Join** step to specify its properties (Figure 30). Set **First step** as **Sort rows**, **Second step** as **Sort rows 2**, and **Join Type** as **INNER**. Click both of the "**Get key fields**" at left and right to get the possible fields to join. In the left table, delete other fields except Day, Month and Year fields. In the right table, delete other fields except *TIMEDAY*, *TIMEMONTH*, and *TIMEYEAR* fields. Then click OK.

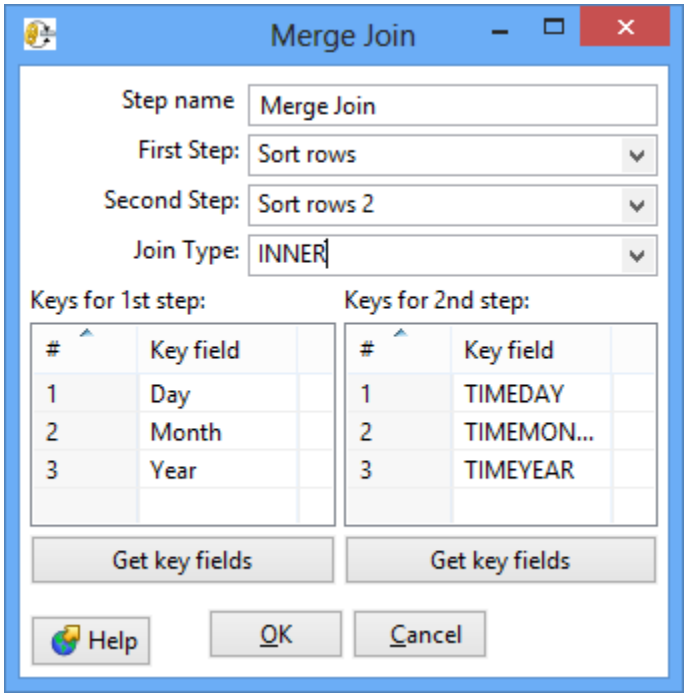
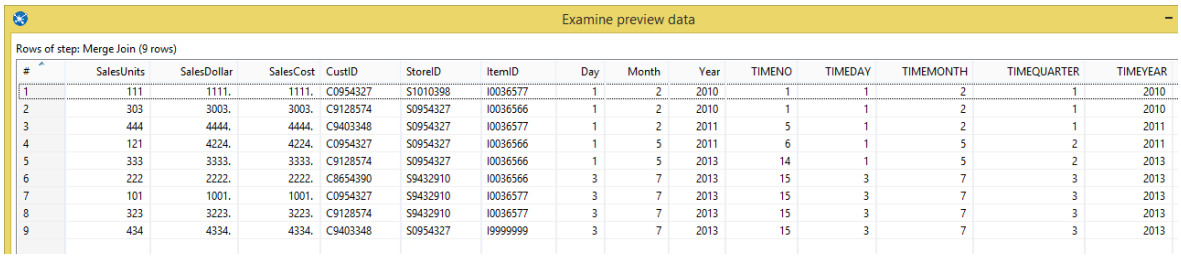


Figure 30: Property Edit Window of Merge Join Node

Step 6.10 Now, we have finished the inner join between Excel input and *SSTimeDim* table.

Step 6.11 Click **Run this transformation** icon  again. Click the Quick **Launch** button to view the output of the transformation. Figure 31 shows the output of **Merge join** coming from two sort rows**.** You will notice that **ONLY 9** rows combined found match out of **10** from **Sort rows**; out of **16** from **Sort rows 2**. Click Close buttons to exit preview data.


Examine preview data

Rows of step: Merge Join (9 rows)

#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year	TIMENO	TIMEDAY	TIMEMONTH	TIMEQUARTER	TIMEYEAR
1	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010	1	1	2	1	2010
2	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010	1	1	2	1	2010
3	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011	5	1	2	1	2011
4	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011	6	1	5	2	2011
5	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013	14	1	5	2	2013
6	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013	15	3	7	3	2013
7	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013	15	3	7	3	2013
8	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013	15	3	7	3	2013
9	434	4334.	4334.	C9403348	S0954327	I9999999	3	7	2013	15	3	7	3	2013

Figure 31: Transformation output of Merge Join Node

Step 7 – Inner join the *SSItem*, *SSCustomer*, and *SSStore* tables.

Similar to getting data from the *SSTimeDim* table in the previous section, inner joining these tables requires **Table Input** components. First, we set the connection and query properties for the *SSItem* table. Note that these tables should exist in your DB2 schema (STAR) before these steps.

Step 7.1 Drag and drop the **Table Input 2** into the design pane.

Step 7.2 Double click on the newly created component to open its Basic Settings pane. Specify the connection as shown in the previous figure.

Step 7.3 Use “SSItem” as the Table Name value and “SELECT * FROM SSItem” as the Query value.

Step 7.4 Create two **sort row** components: **Sort rows 3** and **Sort rows 4**, connecting **Merge Join** and **SSItem** respectively. See the field to be sorted as: **ItemID** and **ITEMID** respectively.

Step 7.5 Drag and drop the **Merge Join 2** into the design pane. Connect **Sort rows 3** and **Sort rows 4** to **Merge Join 2**. Set the field to be joined as **Item ID** and **ITEMID**.

Step 7.6 Check global view of all nodes and connections after all steps in Step 7 is shown by Figure 32.

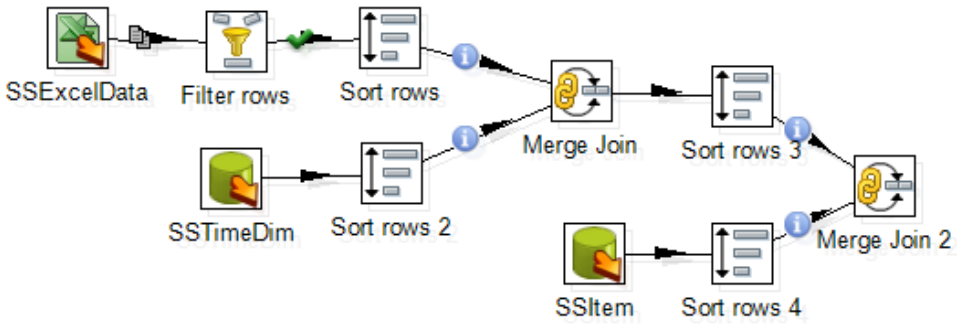


Figure 32: Global View of All Nodes and Connections after Step 7

Step 7.7 View transformation output of **Merge Join 2** at Figure 33.

Examine preview data

Rows of step: Merge Join 2 (8 rows)

#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year	TIMENO	TIMEDAY	TIMEMONTH
1	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010	1	1	2
2	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011	6	1	5
3	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013	14	1	5
4	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013	15	3	7
5	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010	1	1	2
6	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011	5	1	2
7	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013	15	3	7
8	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013	15	3	7

Figure 33: 8 rows produced at Merge Join 2

Step 8 – Inner join the remaining tables.

Step 8.1 Inner join the tables called *SSCustomer* and *SSStore* in your transformation using the same method described previously.

Step 8.2 For the *SSCustomer* step, connect the *CustID* (from Excel file) and CUSTID (from Database) fields. Figure 34 is the output of **Merge Join 3**.

Examine preview data

Rows of step: Merge Join 3 (8 rows)

#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year	TIMENO	TIMEDAY	TIMEMONTH	TIMEQUARTER	TIMEYEAR
1	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011	6	1	5	2	2011
2	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010	1	1	2	1	2010
3	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013	15	3	7	3	2013
4	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013	15	3	7	3	2013
5	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010	1	1	2	1	2010
6	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013	14	1	5	2	2013
7	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013	15	3	7	3	2013
8	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011	5	1	2	1	2011

Figure 34: CustID is sorted in Merge Join 3

Step 8.3 For the *SSStore* step, connect the *StoreID* (from Excel file) and STOREID (from Database) fields. Figure 35 is the output of **Merge Join 4**.

Examine preview data

Rows of step: Merge Join 4 (8 rows)

#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year	TIMENO	TIMEDAY	TIMEMONTH	TIMEQUARTER	TIMEYEAR
1	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011	6	1	5	2	2011
2	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010	1	1	2	1	2010
3	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013	14	1	5	2	2013
4	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011	5	1	2	1	2011
5	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010	1	1	2	1	2010
6	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013	15	3	7	3	2013
7	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013	15	3	7	3	2013
8	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013	15	3	7	3	2013

Figure 35: Transformation output of Merge Join 4 is sorted by StoreID

Step 8.4 Check global view of all nodes and connections after Step 8 is shown by Figure 36.

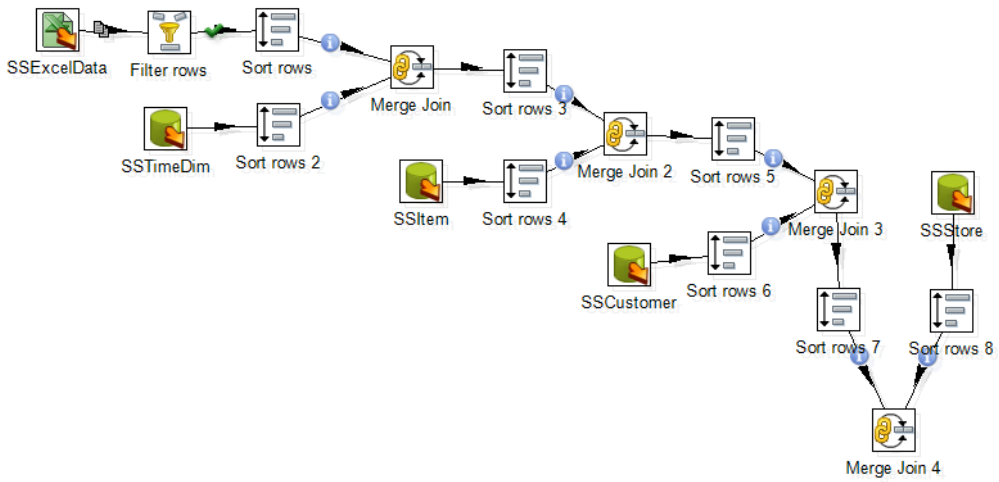


Figure 36: Global View of All Nodes and Connections after Step 8

Step 9 – Create and connect an Add Sequence step to generate values for the *SalesNo* column.

Step 9.1 Under the **Design** tab, expand the contents of the **Transform** node.

Step 9.2 Click and drag an **Add sequence** step into your transformation; create a hop between the **Merge Join 4** and **Add Sequence** steps (Figure 37).

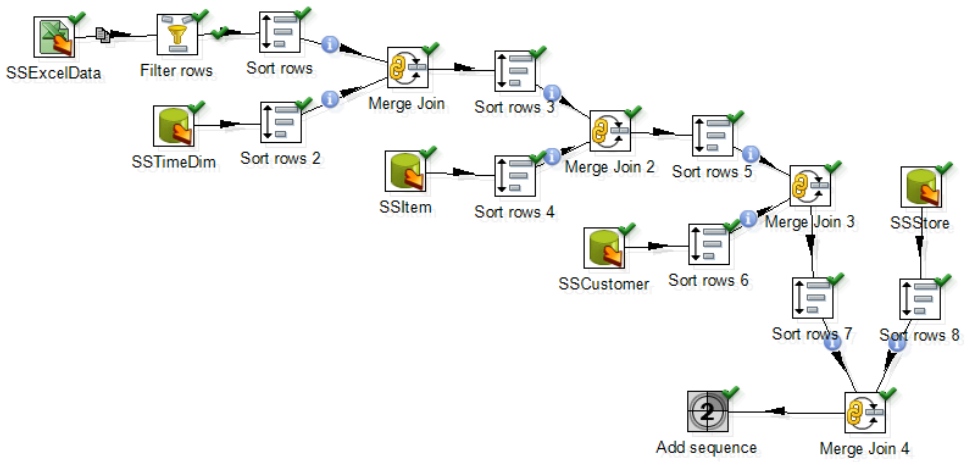


Figure 37: Global View of All Nodes and Connections after Step 4

Step 9.3 Double click on the newly created component to open its Basic Settings pane.
Step 9.4 Set **SalesNo** as the name of value. Check the box for use DB to get sequence. Connection is still set as **DB2Connection.** Set **SSSalesNoSeq** as sequence name (Figure 38).

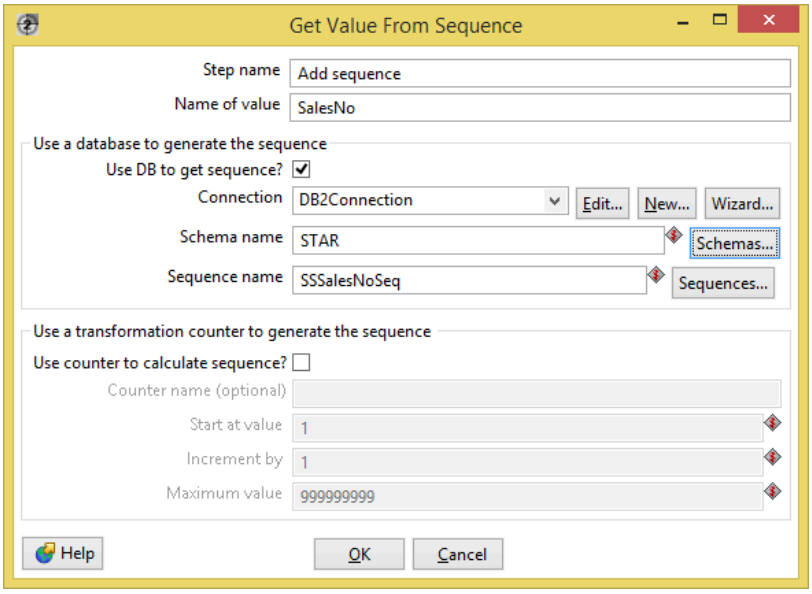


Figure 38: Property Edit Window of Add sequence node

IMPORTANT NOTE: Do not *Run transformation* for **Add Sequence.** Running of this sequence should be done **ONCE** as the final output in loading of correct data. You will generate SalesNo values that won't match the output at the end of part 1 exercise.

5. Insert data into the SSSales table

Step 1 - Under the **Design** tab, expand the contents of the **Output** node.

Step 2 - Click and drag an **Insert/Update** step into your transformation; create a hop between the **Add sequence** and **Insert/Update** steps. Figure 39 shows the **Insert/Update** node (set name as SSSales) connected to **Add sequence** Node.

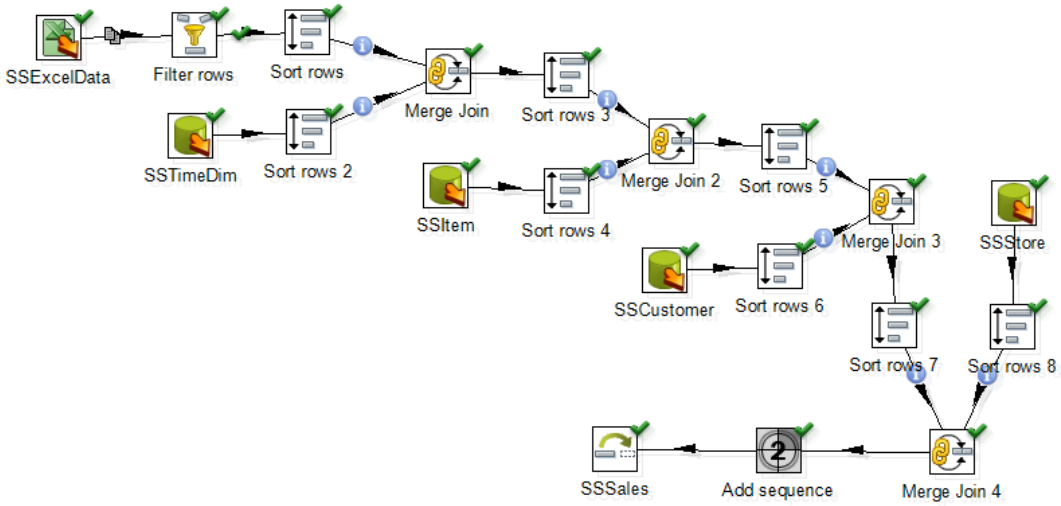


Figure 39: Connect Insert/Update Node to Last Merge Join Node

Step 3 - Double click the **Insert/Update** component, to specify its properties. Set the **step name** as **SSSales**. Connection name is still **DB2Connection**. Type in the **Target schema** as <user-id schema>. **Target table** as **SSSales**. **DON'T** click the button "**Get fields**". Instead, select the names from the two table fields and set the comparator between them to "**=**". See Figure 40.

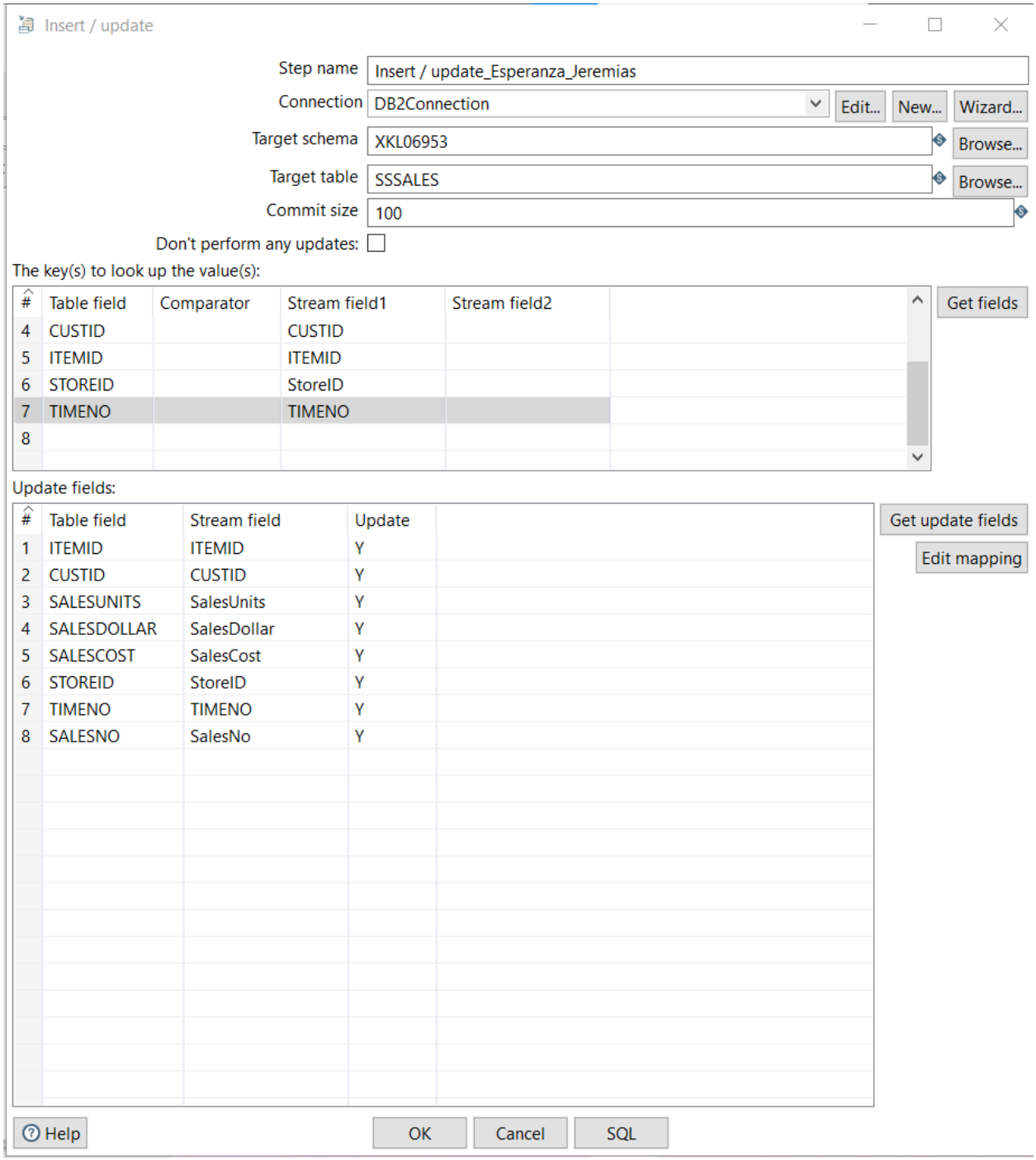


Figure 40: Updating of Insert/Update Node

Step 4 - Click the button "**Get Updated fields**".

Important Note: There is a bug in the program that will create error. To avoid the unwanted error messages, click **OK**. Double click again **SSSales** node to refresh/reload and continue editing the window.

Step 5 - Click on the "**Edit mapping**" button to edit mapping. The mapping edit window is shown by Figure 41. Check if the fields match under **Mappings** as indicated in the figure.

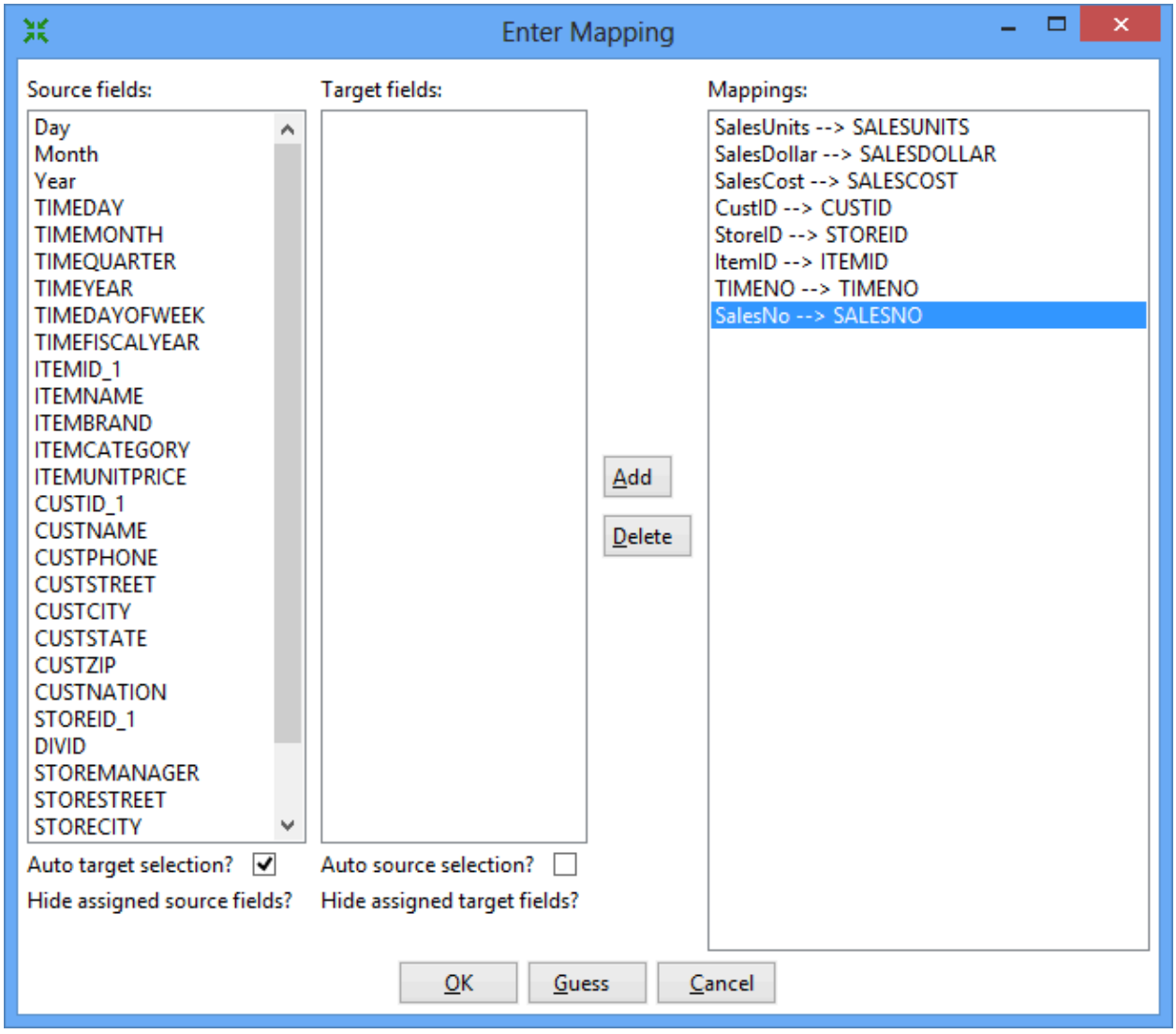
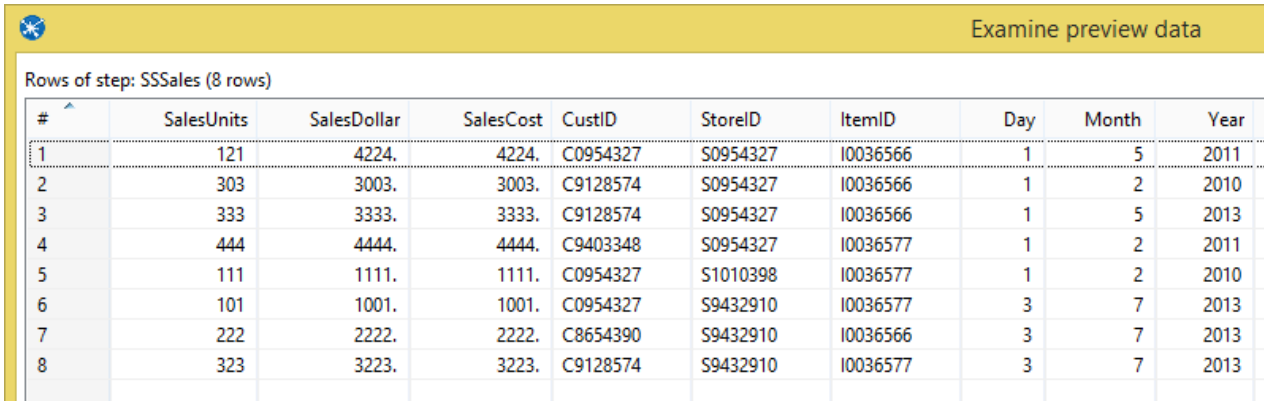


Figure 41: Mapping Edit Window

Step 6 - Click **OK** buttons to finish editing the window. Select the **SSSales** step and run a preview by clicking on . In the transformation debug dialog click on **Quick Launch**.

Step 7 - The Examine preview data window is displayed by Figure 42.

IMPORTANT NOTE: Do not *Run transformation* for **SSSales** step several times**.** Running of this step should be done **ONCE** as the final output in loading of correct data. You will generate incrementing SalesNo values with the same batch of uploaded data.



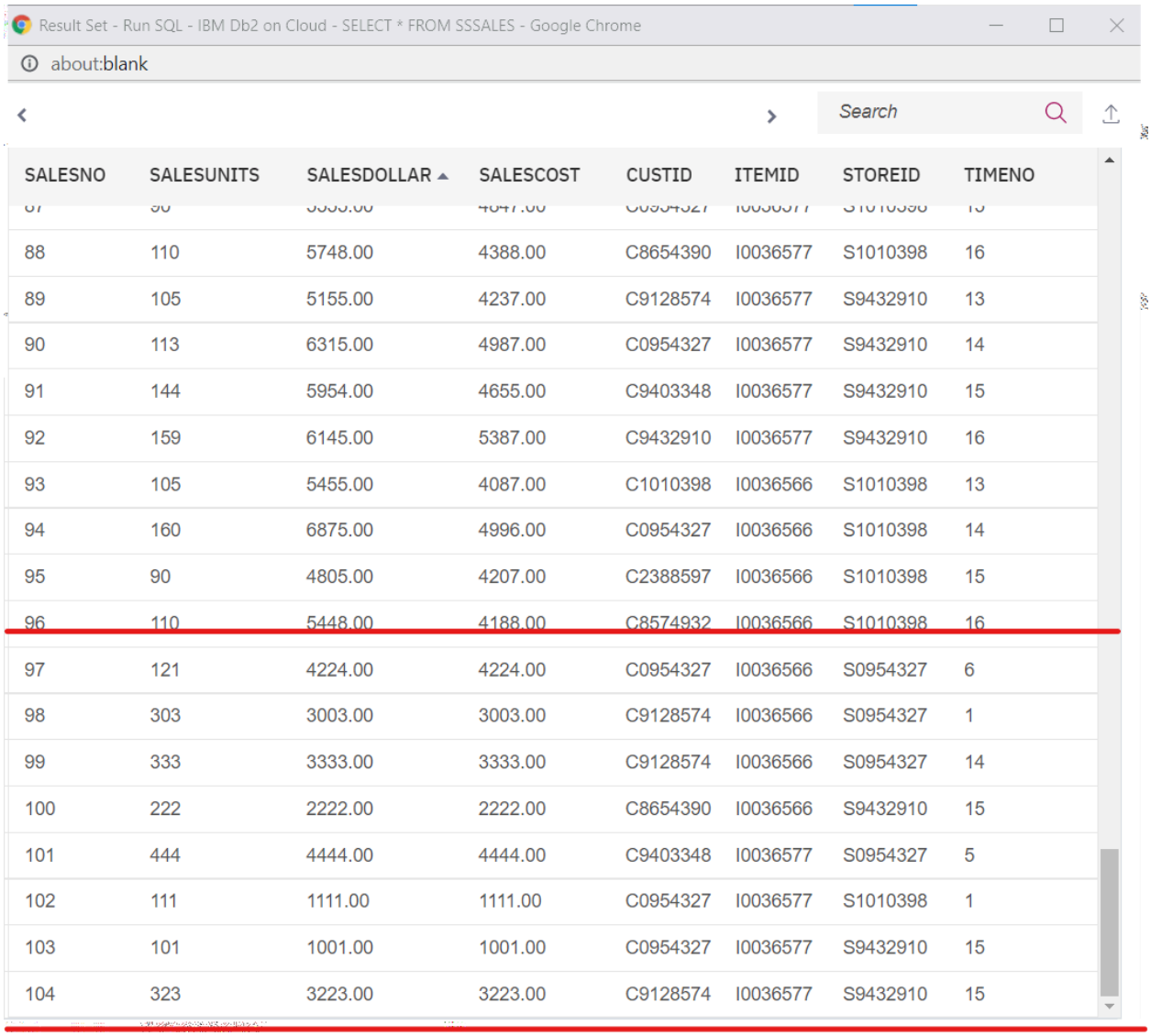
#	SalesUnits	SalesDollar	SalesCost	CustID	StoreID	ItemID	Day	Month	Year
1	121	4224.	4224.	C0954327	S0954327	I0036566	1	5	2011
2	303	3003.	3003.	C9128574	S0954327	I0036566	1	2	2010
3	333	3333.	3333.	C9128574	S0954327	I0036566	1	5	2013
4	444	4444.	4444.	C9403348	S0954327	I0036577	1	2	2011
5	111	1111.	1111.	C0954327	S1010398	I0036577	1	2	2010
6	101	1001.	1001.	C0954327	S9432910	I0036577	3	7	2013
7	222	2222.	2222.	C8654390	S9432910	I0036566	3	7	2013
8	323	3223.	3223.	C9128574	S9432910	I0036577	3	7	2013

Figure 42: Execution Report Window

Step 8 – Verify SSSales if updated in DB2 Cloud:

- SELECT SalesNo, SalesUnits, TimeNo, CustID, StoreID, ItemID, SalesUnits, SalesCost FROM SSSALES

You should see 104 rows with 8 new rows added to the 96 rows of SSSales table. (Figure 43).



SALESNO	SALESUNITS	SALESDOLLAR	SALESCOST	CUSTID	ITEMID	STOREID	TIMENO
88	110	5748.00	4388.00	C8654390	I0036577	S1010398	16
89	105	5155.00	4237.00	C9128574	I0036577	S9432910	13
90	113	6315.00	4987.00	C0954327	I0036577	S9432910	14
91	144	5954.00	4655.00	C9403348	I0036577	S9432910	15
92	159	6145.00	5387.00	C9432910	I0036577	S9432910	16
93	105	5455.00	4087.00	C1010398	I0036566	S1010398	13
94	160	6875.00	4996.00	C0954327	I0036566	S1010398	14
95	90	4805.00	4207.00	C2388597	I0036566	S1010398	15
96	110	5448.00	4188.00	C8574932	I0036566	S1010398	16
97	121	4224.00	4224.00	C0954327	I0036566	S0954327	6
98	303	3003.00	3003.00	C9128574	I0036566	S0954327	1
99	333	3333.00	3333.00	C9128574	I0036566	S0954327	14
100	222	2222.00	2222.00	C8654390	I0036566	S9432910	15
101	444	4444.00	4444.00	C9403348	I0036577	S0954327	5
102	111	1111.00	1111.00	C0954327	I0036577	S1010398	1
103	101	1001.00	1001.00	C0954327	I0036577	S9432910	15
104	323	3223.00	3223.00	C9128574	I0036577	S9432910	15

Figure 43: Inserted Data in DB2 SSSales Table coming from Excel input file

Step 9 - If you do not see the extra rows, the DB2 output component had a failure. To see the error, check the **Execution Results** section.

Part 2: MS Access database as data source

6. Load second data source from Access

The next part of the exercise involves creation of a new transformation to process the Microsoft Access data source.

Step 1 - Create new transformation. Click File ⍰ New ⍰ Transformation. A new **Transformation 1** tab is created at the canvas area. See Figure 44.

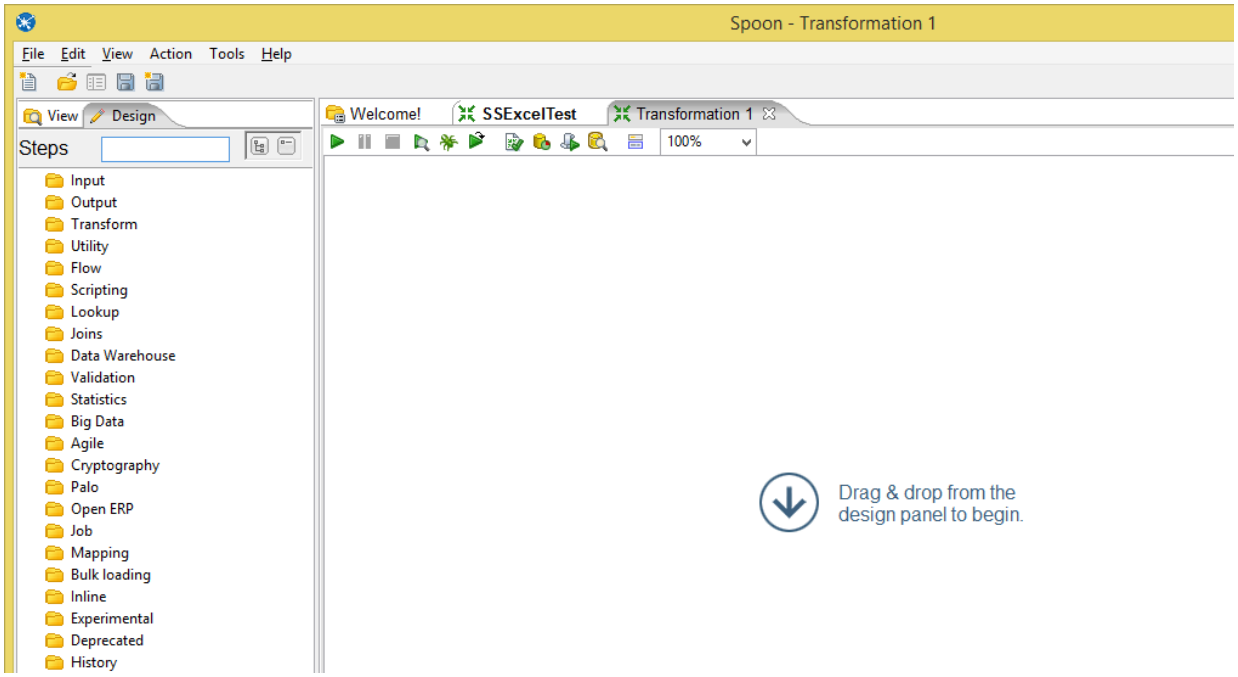


Figure 44. New transformation created

Step 2 – Click **View** tab. Right-click **Transformation 1** ⍰ **Settings**. Set transformation name with SSAccessDBTest. Save the new transformation by clicking **File** ⍰ **Save** with filename **SSAccessDBTest** at *C:\..\Documents*. You will have a revised Window interface set (see Figure 45).

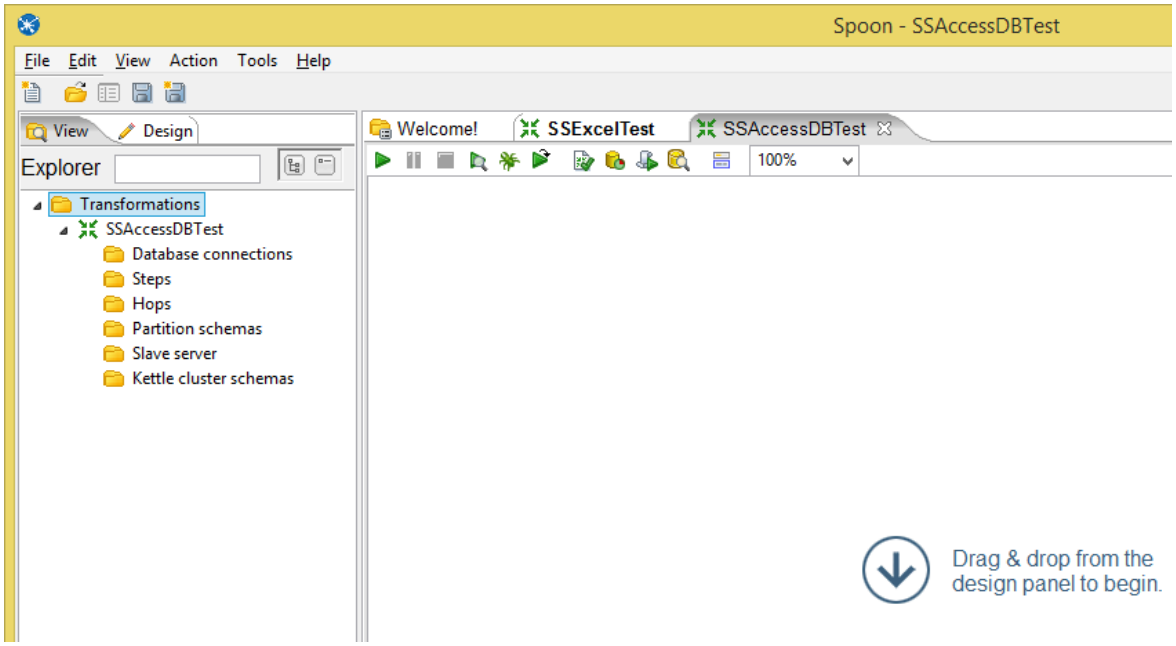


Figure 45: Saved/Set with new transformation name

Step 3- Add the Access Input Step

Step 3.1 Under the **Design** tab, expand the **Input** node.

Step 3.2 Select and drag a **Microsoft Access Input** step onto the canvas on the right (see Figure 46).

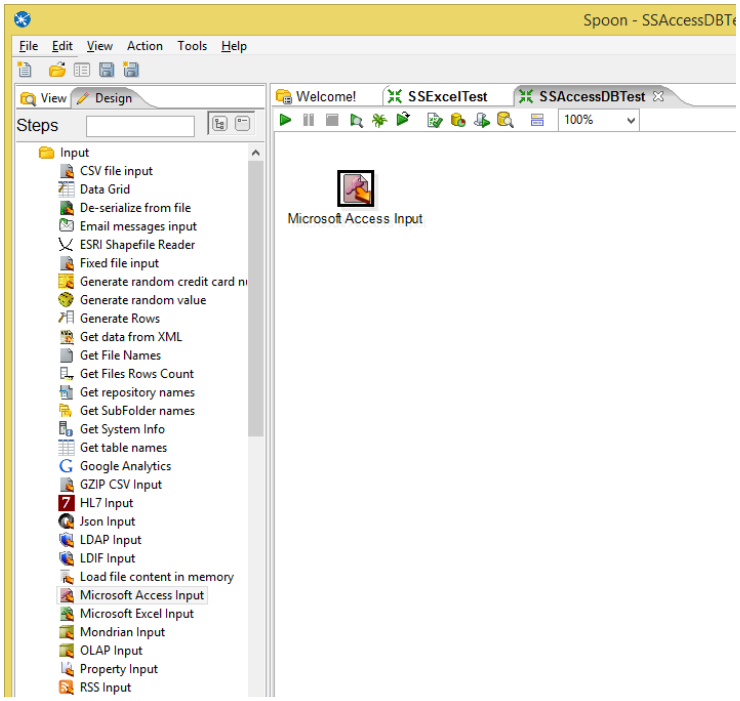


Figure 46: New Microsoft Access Input Node

Step 3.3 Double Click on the **Microsoft Access Input**. The edit properties dialog box associated with the **Microsoft Access Input** step appears (Figure 47). In this dialog box, you specify the properties related to a particular step.

Step 3.4 Set name for the Access Input as **Sales** and specify the Access data source path in the **Files** tab. Click **Browse** and select **SSAccessData.mdb**. Click **Add** to select the file. You should have the window set at Figure 47.

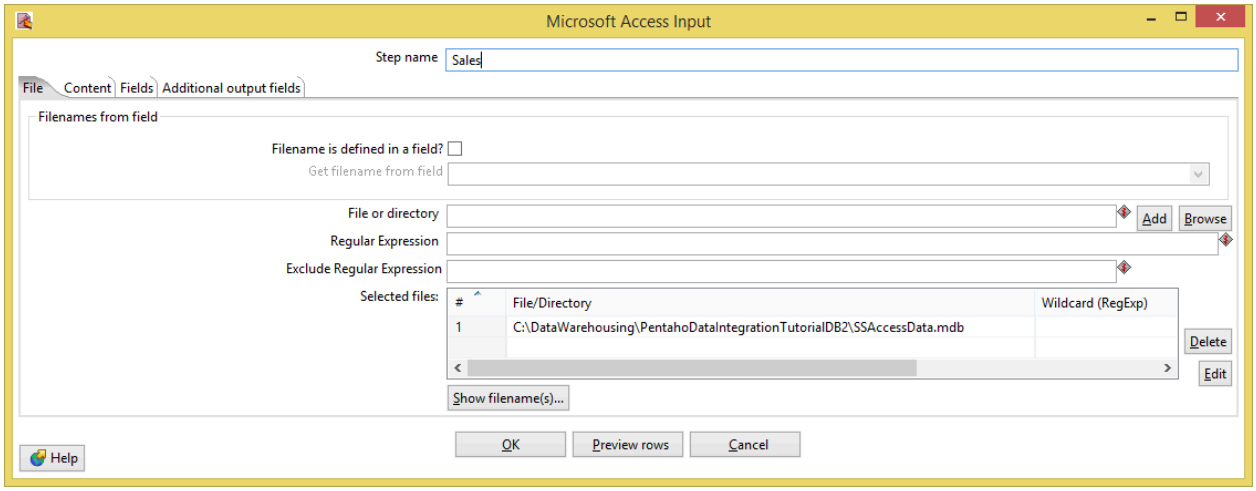


Figure 47: Property Edit Window of Microsoft Access Input Node

Step 3.5 In the tab named **Content**, click the button "**Get tables**" of **table** section. There will appear a window (Figure 48). Select **Sales** as the table name, click **OK**.

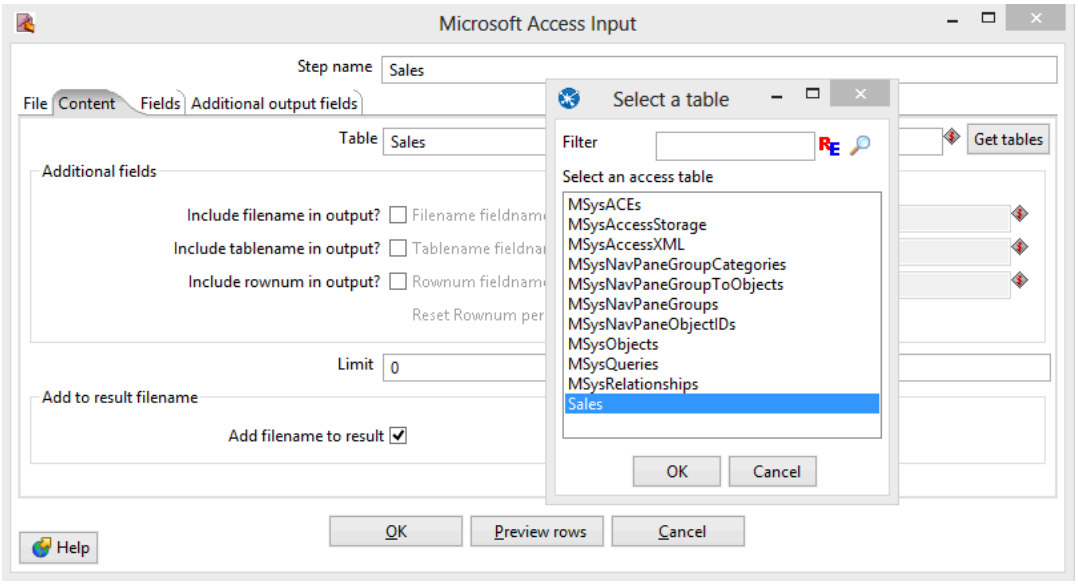


Figure 48: Table Selection Window

Step 3.6 In the tab named **Fields**, click the button "**Get fields**". There will appear a list (Figure 49) showing the fields in the table named **Sales**.

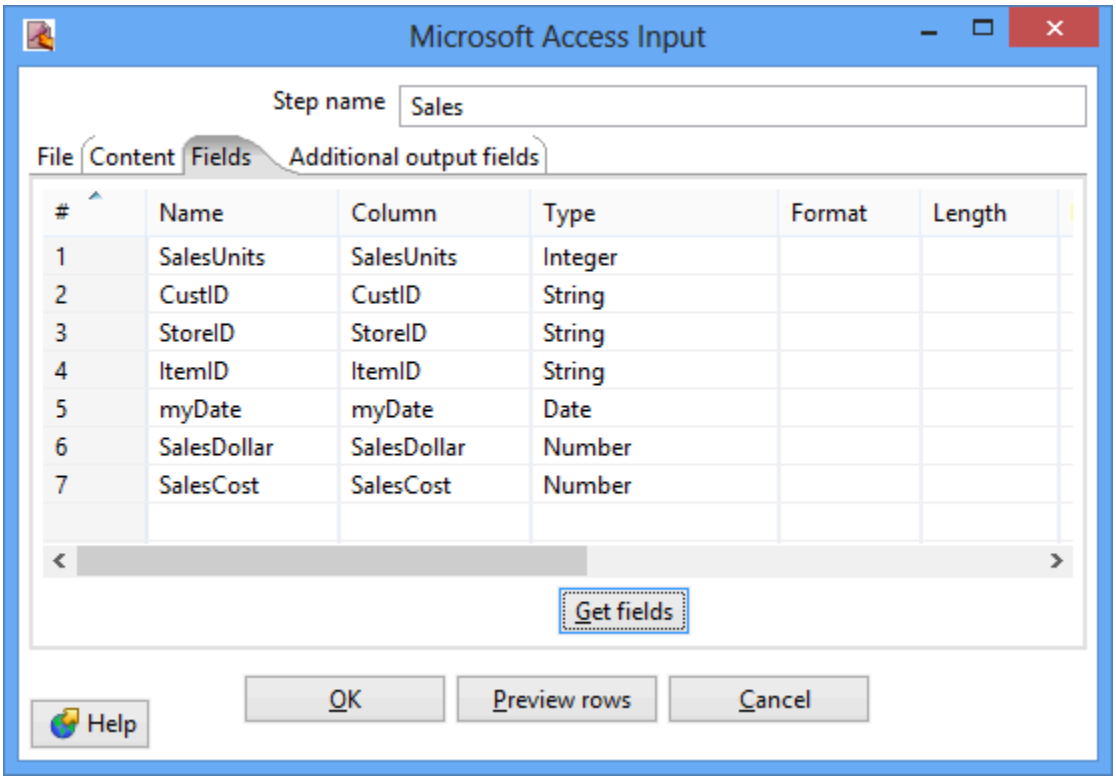


Figure 49: Fields Window for Microsoft Access Input Property Editing

Step 3.7 Click the button "**Preview rows**" to preview the database (Figure 50). When asked for the number of rows type 12 and click OK.

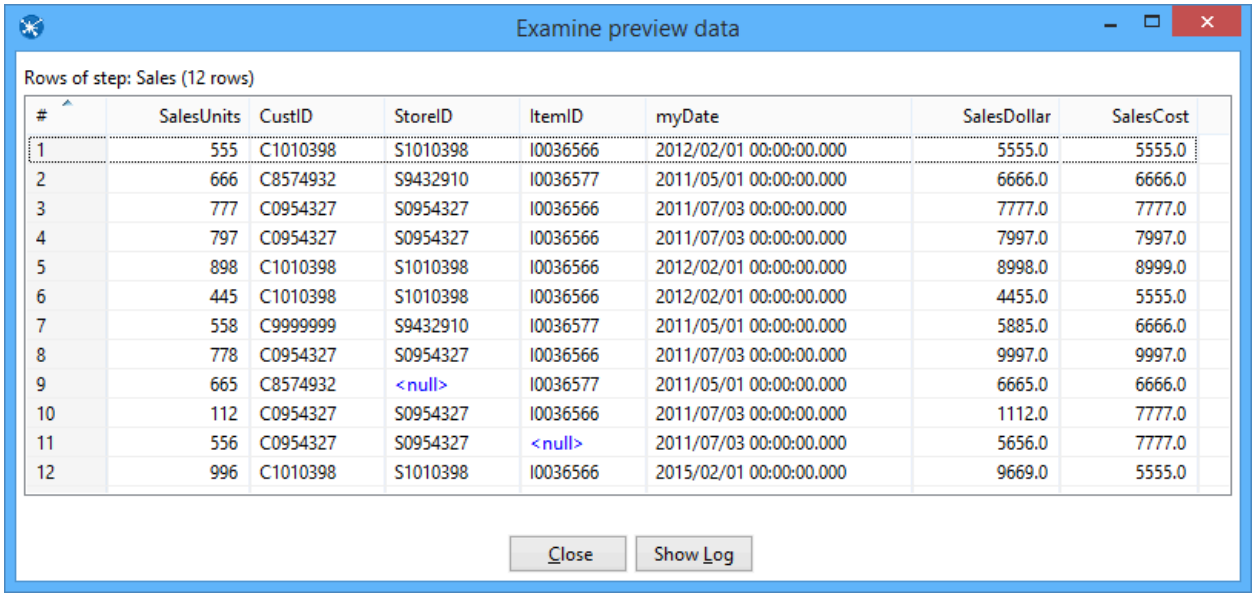


Figure 50: Examine Preview Data Window

Step 3.8 Click **OK** at the bottom of the window. The input icon will change to the shape shown by Figure 51.

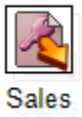


Figure 51: Sales Node Icon

Step 4 –You will add constraint checking for null values using the Filter Rows step.

Step 4.1 Add a Filter Rows step to your transformation. Under the **Design** table, go to **Flow** → **Filter Rows** (Figure 52).

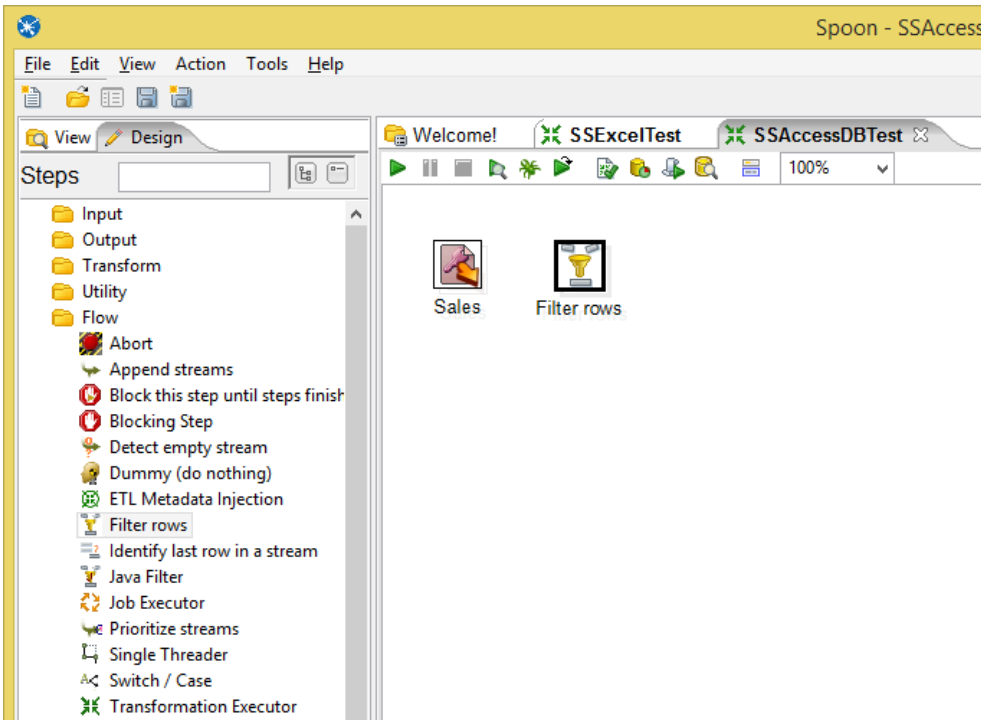


Figure 52: Access Input Node and Filter Node in Spoon

Step 4.2 Create a hop between the **Sales** (Access file input) step and the **Filter Rows** step.

Step 4.3 Double-click the **Filter Rows** step. The **Filter Rows** edit properties dialog box appears.

Step 4.4 In the **Step Name** field type, **Filter rows**.

Step 4.5 Under **The condition**, click <field>. A dialog box that contains the fields you can use to create your condition appears.

Step 4.6 In the **Fields**: dialog box select **SalesUnits** and click **OK**.

Step 4.7 Click on the comparison operator (set to = by default) and select the **IS NOT NULL** function and click **OK**.

Step 4.8 Click the button  , add constraints for other columns. Final window result is at Figure 53.

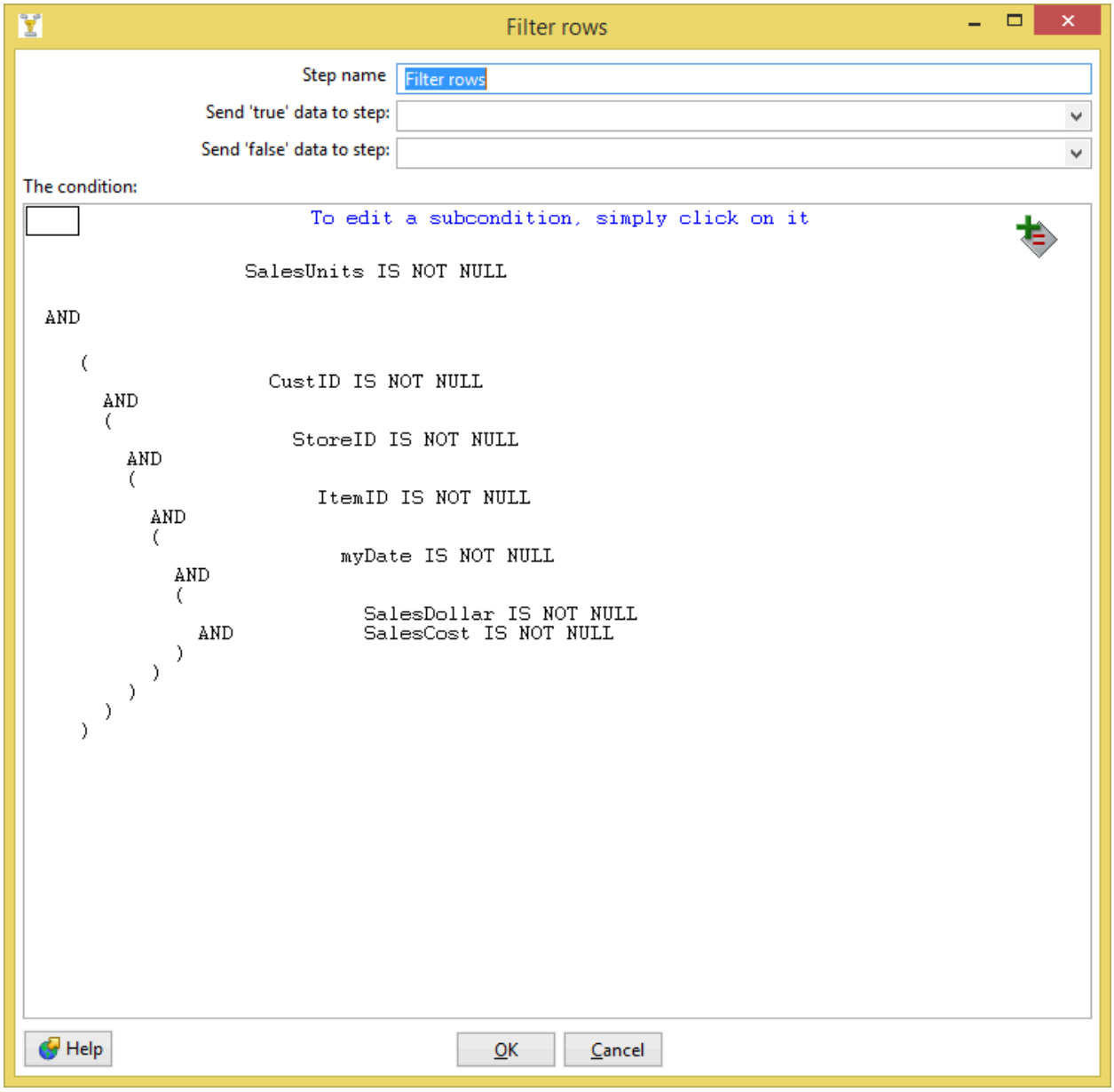


Figure 53: Filter Conditions Window

Step 4.9 Save your transformation.

7. Separate SalesDay fields into Day, Month, Year fields

In this part of the tutorial, you will use the Select Values step to change the format of the myDate field and the Split Fields step to parse the field into date components.

Step 1 - Under the **Design** tab, expand the contents of the **Transform** node.
Step 2 - Click and drag a **Select values** step into your transformation.
Step 3 - Create a "hop" between the **Filter rows** step and the **Select values** step (Figure 54).

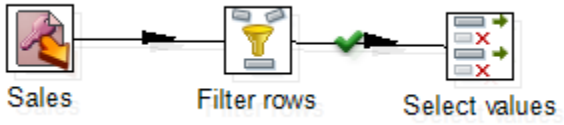


Figure 54: True Filter Results Connected to Select Values Node

Step 4 - Double-click the **Select values** step to open its edit properties dialog box.
Step 5 - In the tab named **Metadata**, click the button "**Get fields to change**", to get the fields to change, which is shown by Figure 55. Change the **Type** of field **myDate** as **String**, change its **Format** as dd-MM-yyyy. Click **OK**.

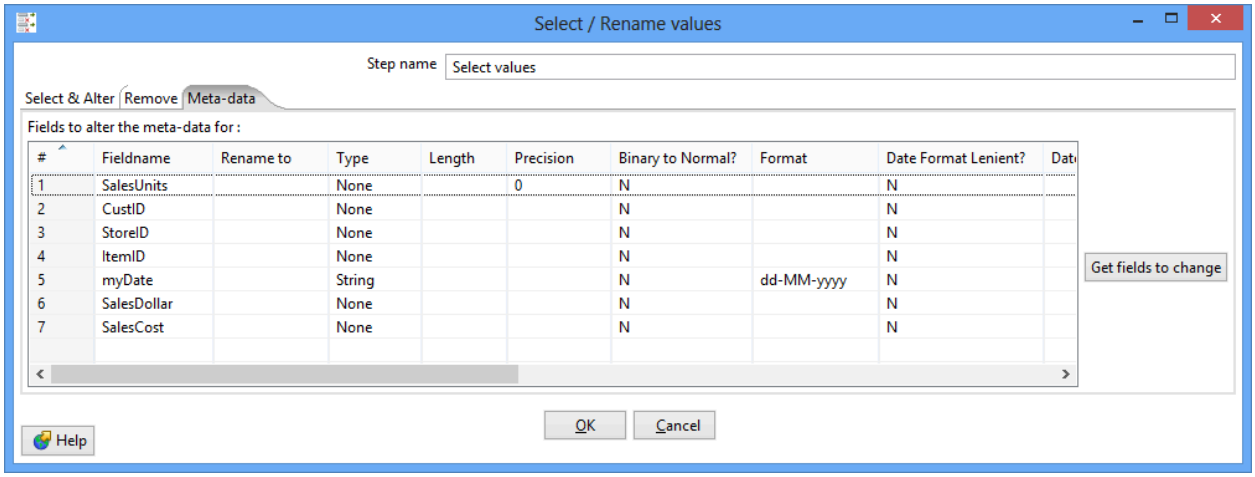


Figure 55: Meta-data Tab of Select Values Property Edit Window

Step 6 - Under the **Design** tab, expand the contents of the **Transform** node.
Step 7 - Click and drag a **Split fields** step into your transformation (Figure 56).



Figure 56: Create Split Fields in Spoon

Step 8 - Create a “hop” between the **Select values** step and the **Split fields** step.

Step 9 - Double-click the **Split fields** step to open its edit properties dialog box (Figure 57).

Step 10 - Select **myDate** in the **Field to split**, type “**-**” as the **Delimiter**. Type in **Year, Month** and **Day** in the Column named **New field**, and set their **Type** as **Number**.

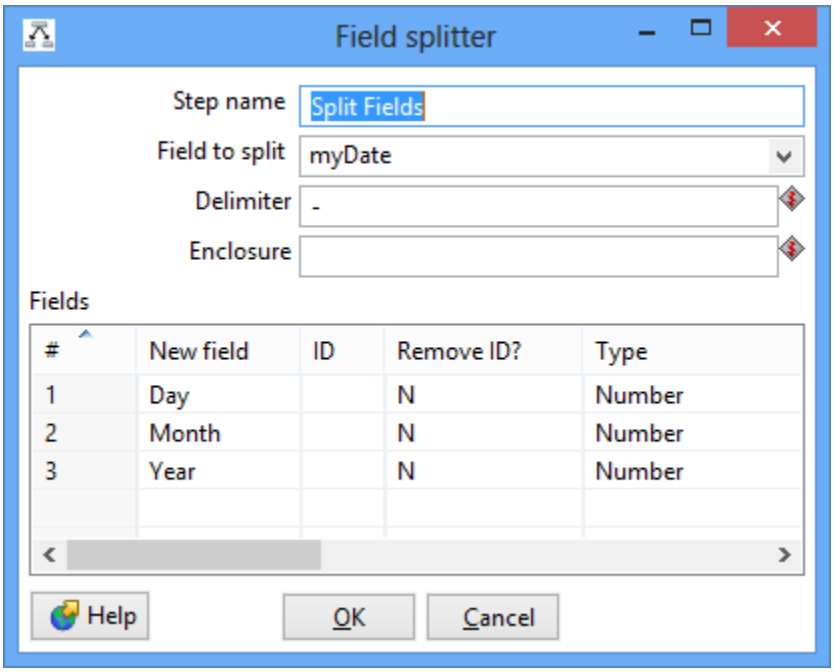
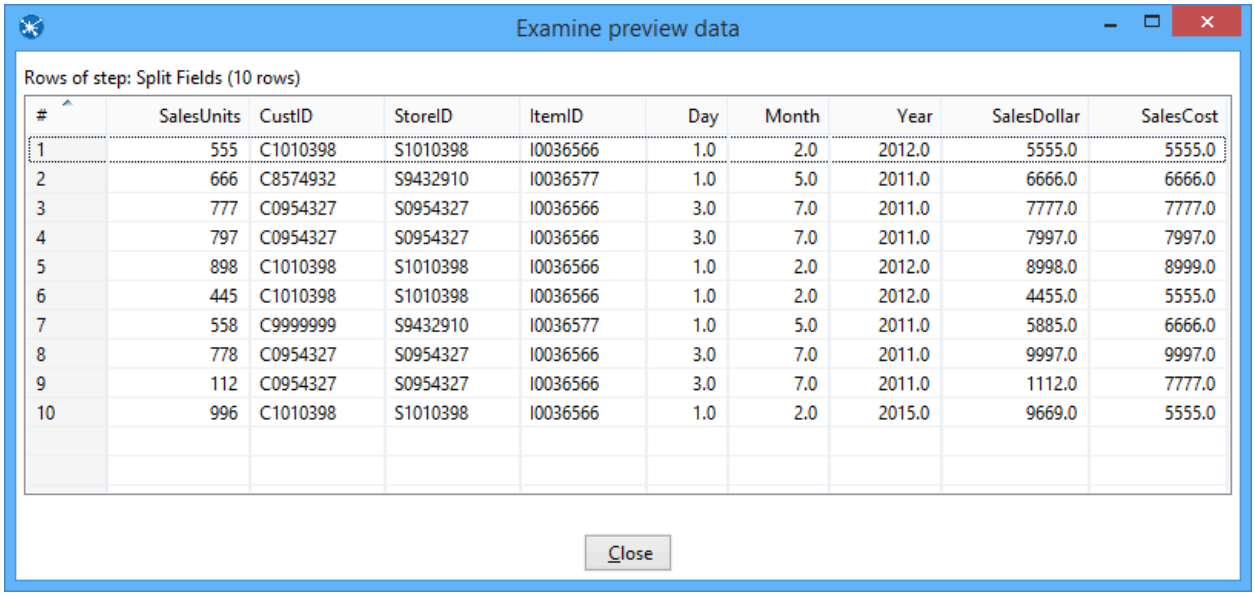


Figure 57: Property Edit Window of Field Splitter Node

Step 11 - Click **OK**.

Step 12 Click  , to preview this transform (Figure 58). Make sure that **Split Fields** step is selected from the left side panel of the transformation debug dialog and click on “**Quick Launch**” button.



#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost
1	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0
2	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0
3	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0
4	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0
5	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0
6	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0
7	558	C9999999	S9432910	I0036577	1.0	5.0	2011.0	5885.0	6666.0
8	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0
9	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0
10	996	C1010398	S1010398	I0036566	1.0	2.0	2015.0	9669.0	5555.0

Figure 48: Examine Preview Data Window

8. Lookup Columns from the DB2 tables

This part of the exercise involves looking up the date from the *SSTimeDim* table to check the validity of dates in the Access data source. In addition, you will lookup primary key columns from other DB2 tables to ensure loaded data does not contain invalid foreign keys. This part of the exercise is similar to Section 3.

Step 1 – Create a database connection.

Step 1.1 Under the **View** tab, double click **Database Connections** node.
Step 1.2 Under the **Database connection** window, set the following properties as seen in Figure 49.

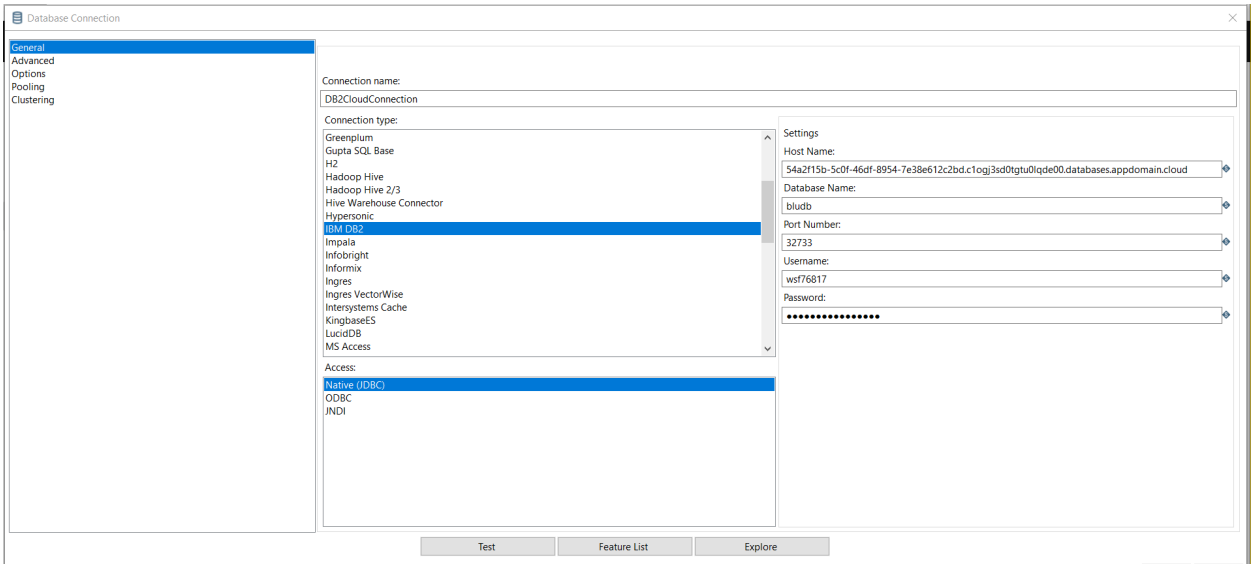


Figure 49: Set database connection property values

Step 1.2.1 - Go to **Options** menu. Supply in the **Parameter** : sslConnection, **Value**: true.

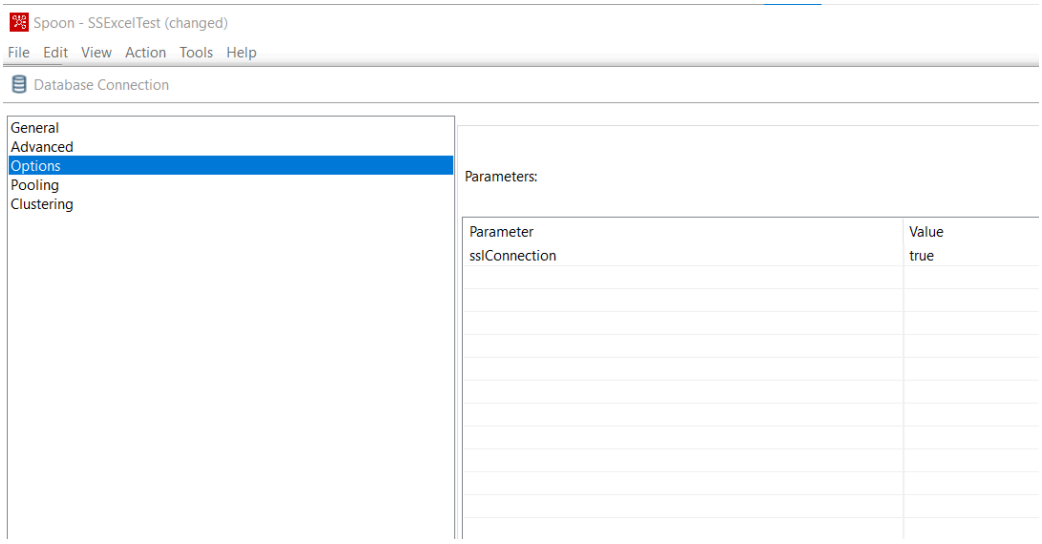


Figure 23.1: Setting the sslConnection to true.

Step 1.3 Click the Test button to test connection. You should achieve Figure 50.

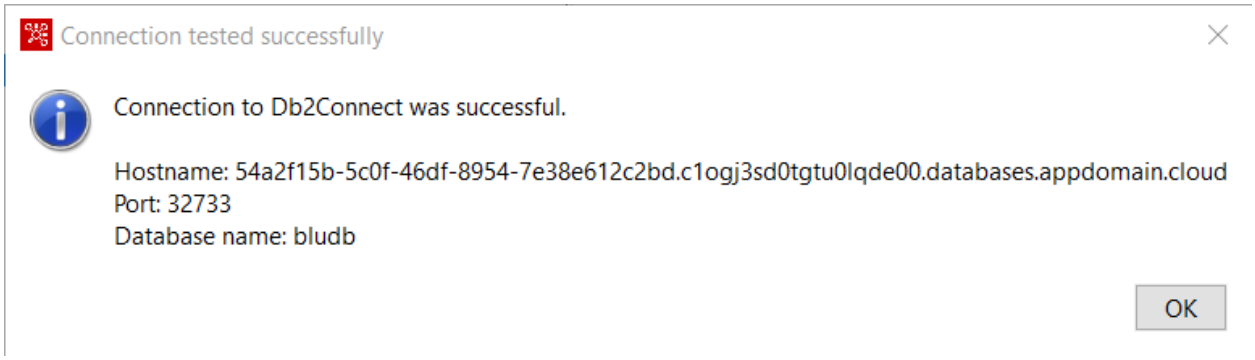


Figure 50: Successful database connection

Step 2 – Access the *SSTimeDim* table from DB2 database.

Step 2.1 Under the **Design** tab, expand the contents of the **Input** node.
Step 2.2 Click and drag a **Table Input** step into your transformation.
Step 2.3 Double-click the **Table Input** step to open its edit properties dialog box.
Step 2.4 Rename your Table Input step to *SSTimeDim*.
Type in "`SELECT * FROM SSTimeDim`" in the SQL section. You can click the **Preview** button to view the database. Click Ok, to exit the Database Connection dialog box.
Step 2.5 Under the **Design** tab, expand the contents of the **Transform** node.
Step 2.6 Click and drag a **Sort Rows** step into your transformation; create a hop between the **Split fields** and **Sort Rows** steps.
Step 2.7 Double-click the **Sort Rows** step to open its edit properties dialog box. Click "**Get fields**" to obtain the fields. Delete other fields except the Day, Month and Year fields. Then click **OK**.
Step 2.8 Add one more sort row component **Sort rows 2**, and a hop connecting the *SSTimeDim* step. In the field specification, delete other fields except *TIMEDAY*, *TIMEMONTH*, *TIMEYEAR* fields.
Step 2.9 Under the **Design** tab, expand the contents of the **Join** node.
Step 2.10 Click and drag a **Merge Join** step into your transformation; create a hop between the **Sort rows, Sort rows 2** and **Merge Join** steps.
Step 2.11 Double-click the **Merge Join** step to specify its properties. Set **First step** as **Sort rows**, **Second step** as **Sort rows 2**, and **Join Type** as **INNER**. Click both of the "**Get key fields**" at left and right to get the possible fields to join. In the left table, delete other fields except Day, Month and Year fields. In the right table, delete other fields except *TIMEDAY*, *TIMEMONTH*, and *TIMEYEAR* fields. Then click **OK**.
Step 2.12 Now, we have finished the inner join between the Access table and *SSTimeDim* table.

Step 2.13 Figure 51 shows the global view of all nodes and connections after Step 2.

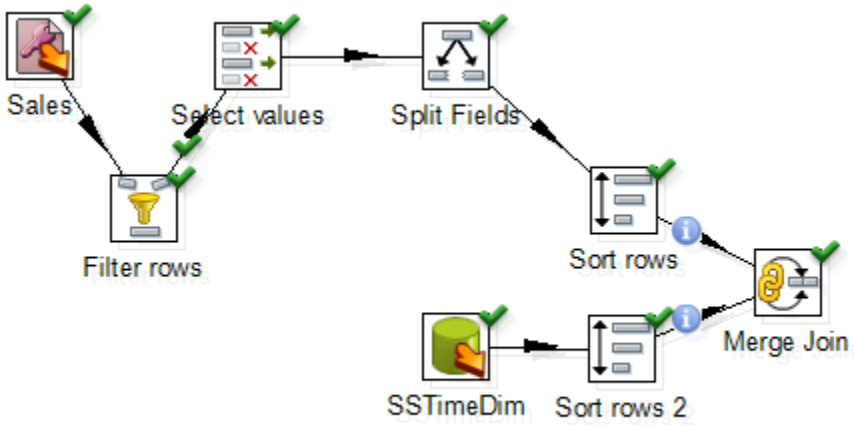


Figure 51: Global View of All Nodes and Connections after Step 2

Step 2.14 Transformation output of **Merge Join** should have output set in Figure 52.

Examine preview data

Rows of step: Merge Join (9 rows)

#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost
1	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0
2	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0
3	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0
4	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0
5	558	C9999999	S9432910	I0036577	1.0	5.0	2011.0	5885.0	6666.0
6	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0
7	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0
8	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0
9	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0

Figure 52: Partial output with 9 rows of Merge Join

Step 3 – Inner join *SSItem*, *SSCustomer*, and *SSStore* to Access table.

Step 3.1 Inner joins the tables named *SSItem*, *SSCustomer*, and *SSStore* in your transformation using the same method described before. Figure 53 will be your guide as the format of node connections.

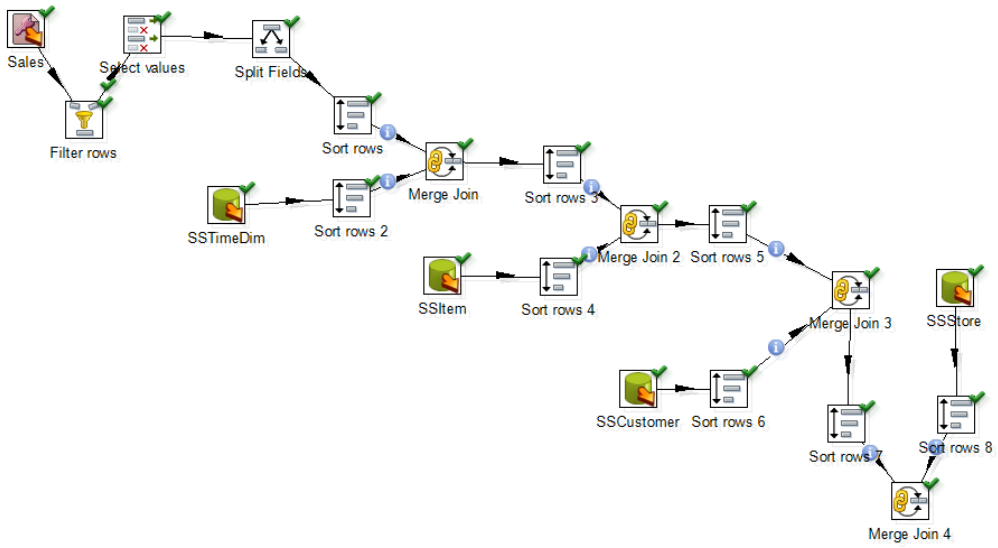


Figure 53: Format of Global View of All Nodes and Connections after Step 3

Step 3.2 For *SSItem* step, connect *ItemID* (from Access file) and *ITEMID* (from Database) fields.

Step 3.3 Figure 54 shows the partial output of **Merge Join 2**.

Examine preview data

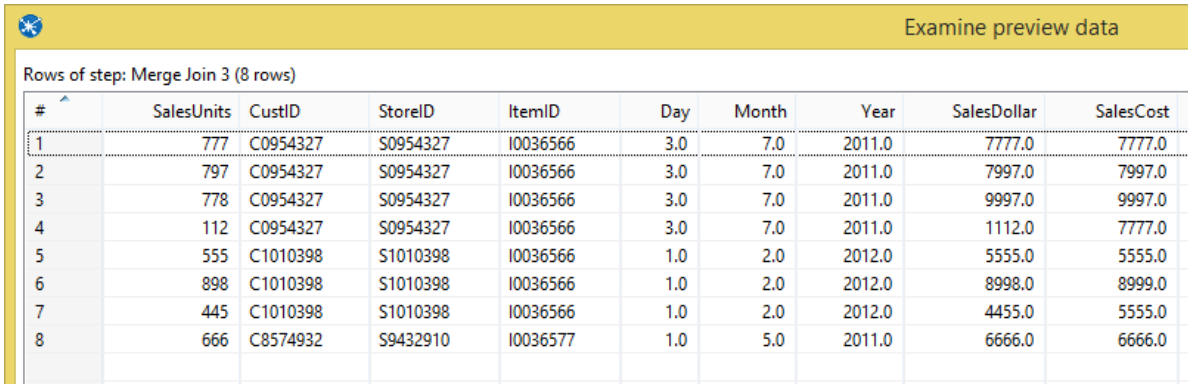
Rows of step: Merge Join 2 (9 rows)

#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost	TIMI
1	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0	
2	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0	
3	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0	
4	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0	
5	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0	
6	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0	
7	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0	
8	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0	
9	558	C9999999	S9432910	I0036577	1.0	5.0	2011.0	5885.0	6666.0	

Figure 54: Partial output with 9 rows of Merge Join 2

Step 3.4 For *SSCustomer* step, connect *CustID* (from Access file) and *CUSTID* (from Database) fields.

Step 3.5 Figure 55 shows the partial output of **Merge Join 3**.


Examine preview data

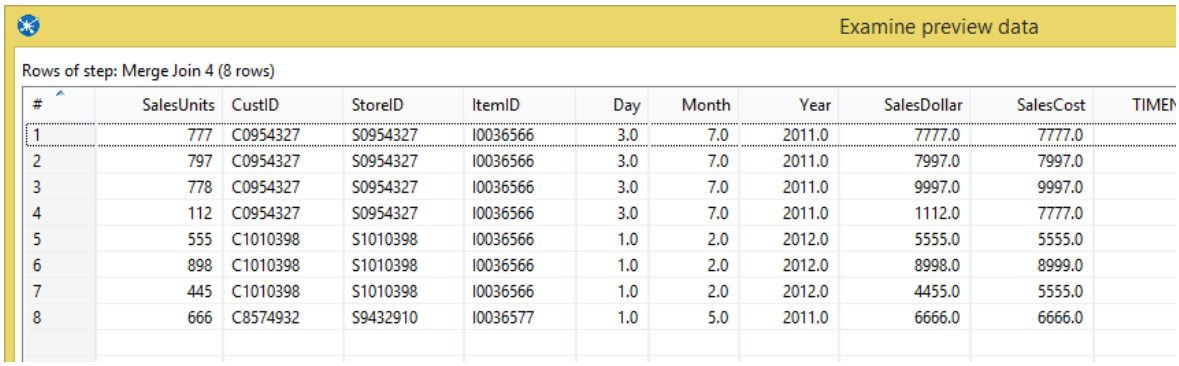
Rows of step: Merge Join 3 (8 rows)

#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost
1	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0
2	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0
3	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0
4	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0
5	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0
6	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0
7	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0
8	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0

Figure 55: Partial output with 8 rows of Merge Join 3

Step 3.6 For *SSStore* step, connect *StoreID* (from Access file) and *STOREID* (from Database) fields.

Step 3.7 Figure 56 shows the partial output of **Merge Join 4**.


Examine preview data

Rows of step: Merge Join 4 (8 rows)

#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost	TIMEN
1	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0	
2	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0	
3	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0	
4	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0	
5	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0	
6	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0	
7	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0	
8	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0	

Figure 56: Partial output with 8 rows of Merge Join 4

Step 4 – Add SalesNo column.

Step 4.1 Under the **Design** tab, expand the contents of the **Transform** node.

Step 4.2 Click and drag **Add sequence** step into your transformation; create a hop between the **Merge Join 4** and **Add Sequence** steps (Figure 57).

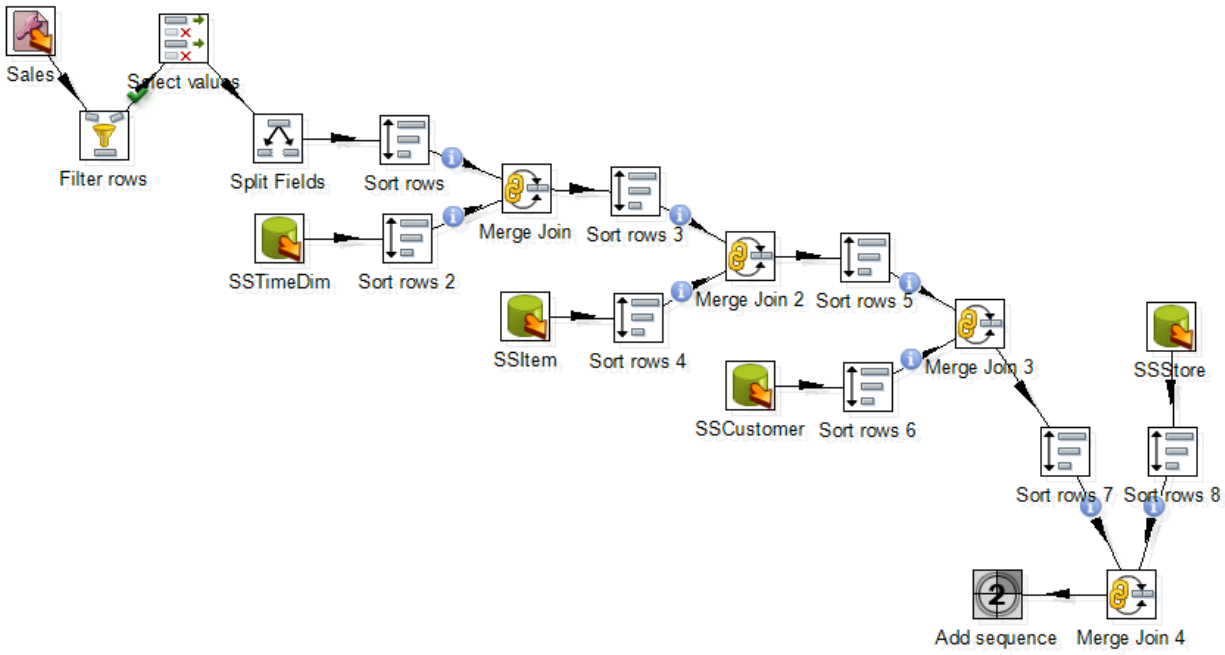


Figure 57: Global View of All Nodes and Connections after Step 4

Step 4.3 Double click on the newly created component to open its Basic Settings pane.
Step 4.4 Set **SalesNo** as the name of value. Check the box for DB **to get the sequence**. **SSSalesNoSeq** as sequence name (Figure 58).

IMPORTANT NOTE: Do not *Run transformation* for **Add Sequence.** Running of this sequence should be done **ONCE** as the final output in loading of correct data. You will generate SalesNo values that won't match the output at the end of Part 2 exercise.

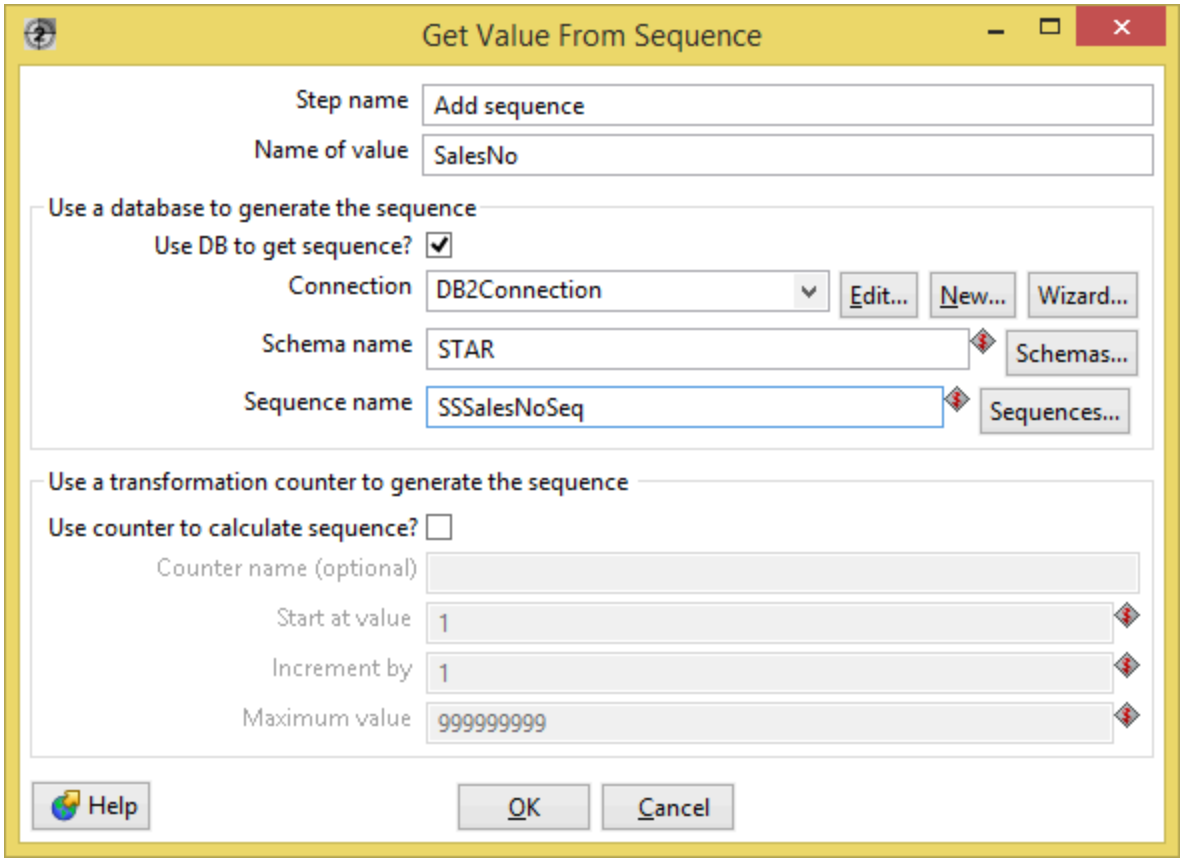


Figure 58: Property Edit Window of Add sequence node

9. Insert data into the SSSales table

Step 1 - Under the **Design** tab, expand the contents of the **Output** node.

Step 2 -Click and drag an **Insert/Update** step into your transformation; create a hop between the **Add sequence** and **Insert/Update** steps. Figure 59 shows the connection.

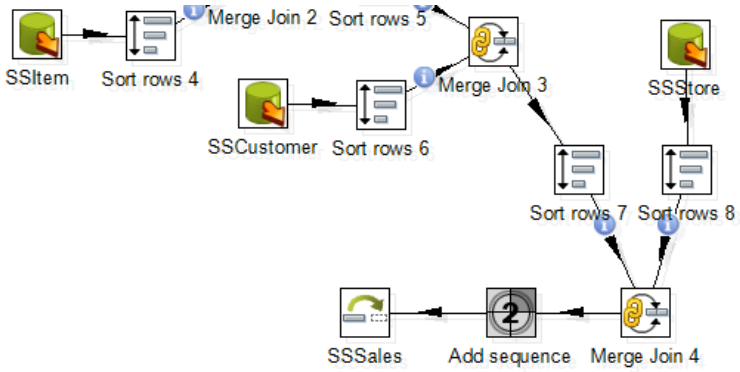


Figure 59: Connect Insert/Update Node (SSSales) to Last Merge Join Node

Step 3 - Double click the **Insert/Update** component, to specify its properties. Set the **step name** as **SSSales**. Type in the **Target table** as **SSSales**. **DON'T** click the buttons "**Get fields**". Instead, select the names from the two table fields and set the comparator between them to "**=**". The partial completed window should look like Figure 60.

Important note: You should have 8 (not 7 as shown) **Table field** and **Stream field1** matching between columns.

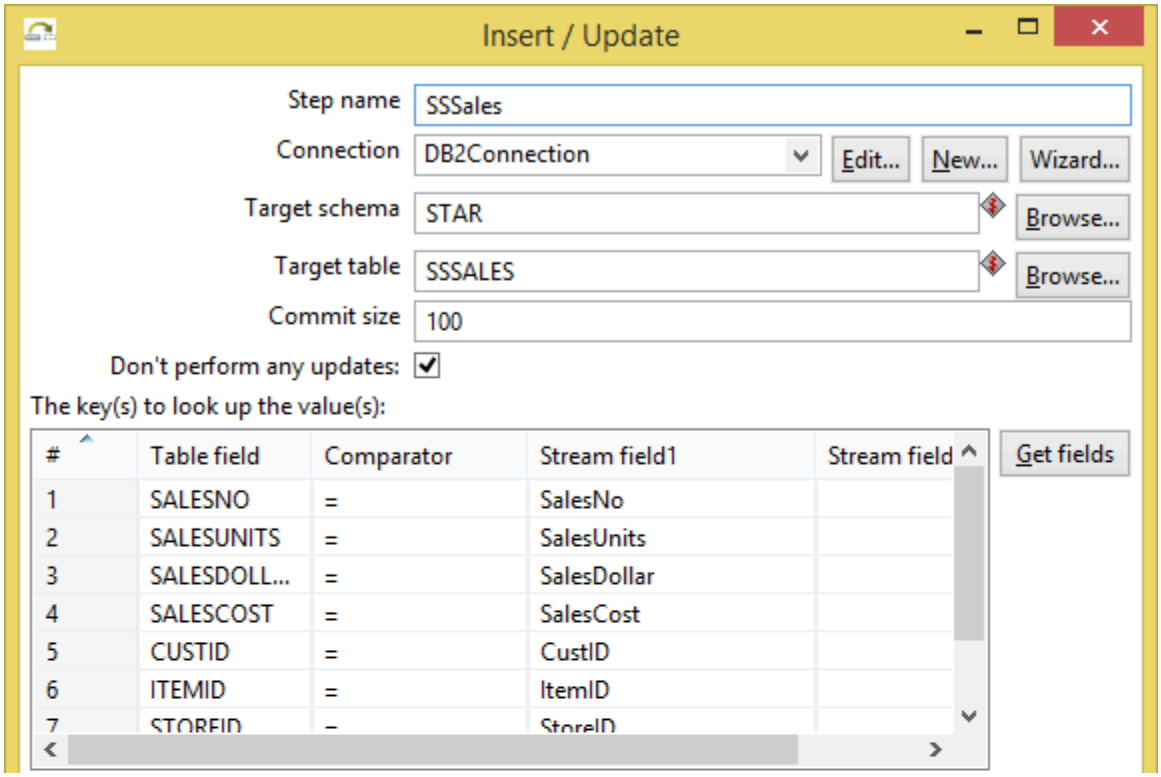


Figure 60: Partially filled-up window for SSSales node

Step 4 - Click the button "**Get Updated fields**".

Important Note: There is a bug in the program that will create error. To avoid the unwanted error messages, click **OK**. Double click again **SSSales** node to refresh/reload and continue editing the window.

Step 5 - Click on "**Edit mapping**" button to edit mapping. The mapping edit window is shown by Figure 61. Check if the fields match under **Mappings** as indicated in the figure.

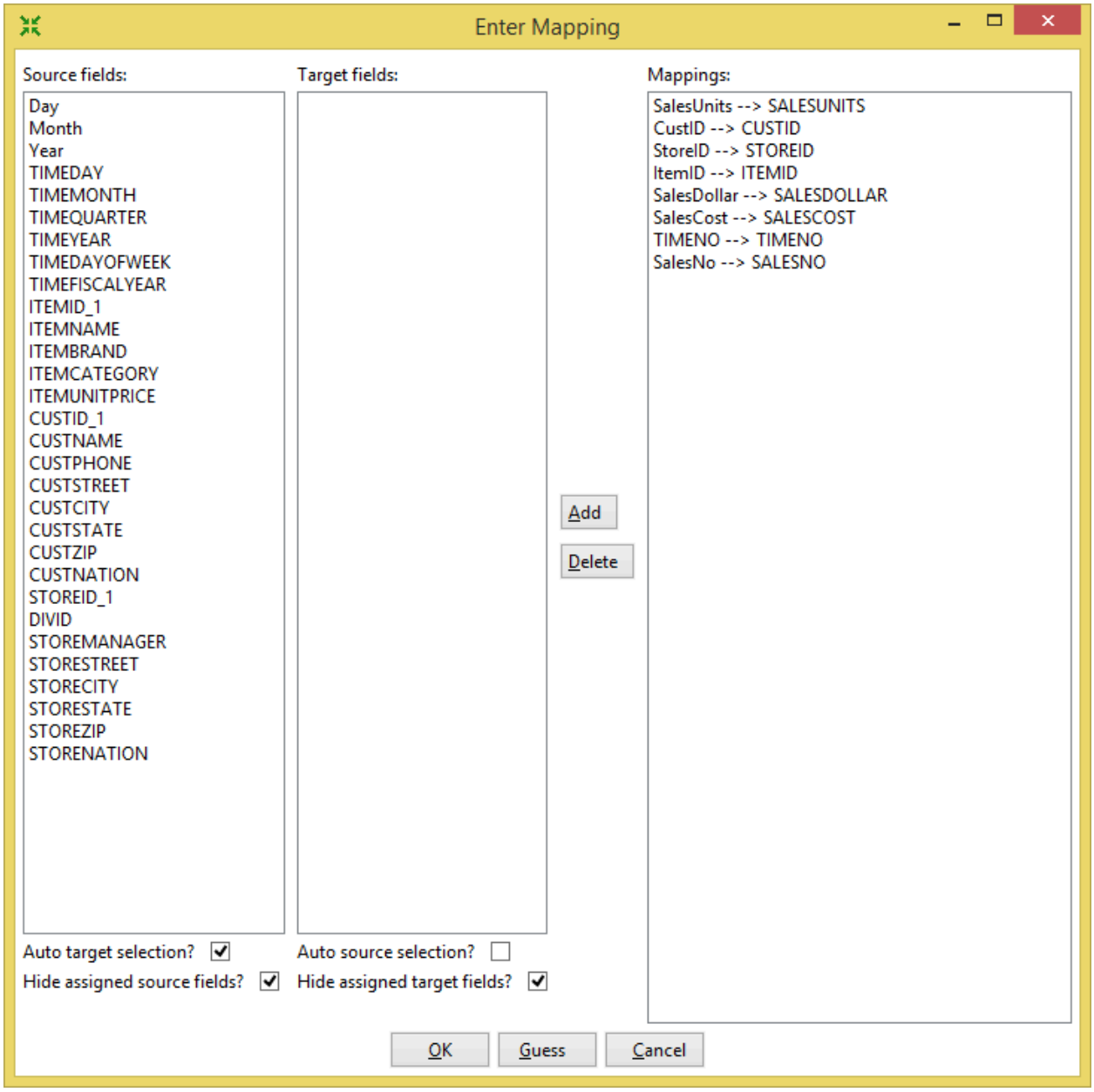
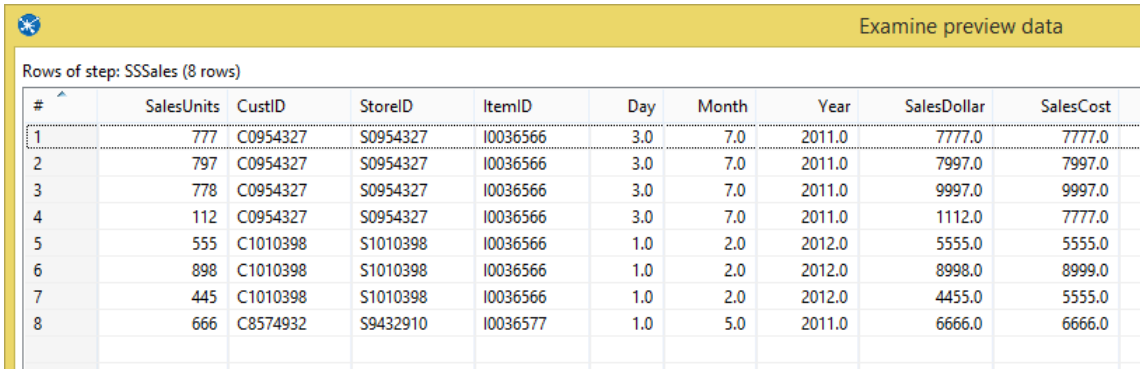


Figure 61: Correct matching of fields takes place at Mapping

Step 6 - Click **OK** buttons to finish editing the window. Select the **SSSales** step and run a preview by clicking on . In the transformation debug dialog click on **Quick Launch**.

Step 7 - The Examine preview data window is displayed by Figure 62.

IMPORTANT NOTE: Do not *Run transformation* for **SSSales** step several times**.** Running of this step should be done **ONCE** as the final output in loading of correct data. You will generate incrementing SalesNo values with the same batch of uploaded data.



Examine preview data

Rows of step: SSSales (8 rows)

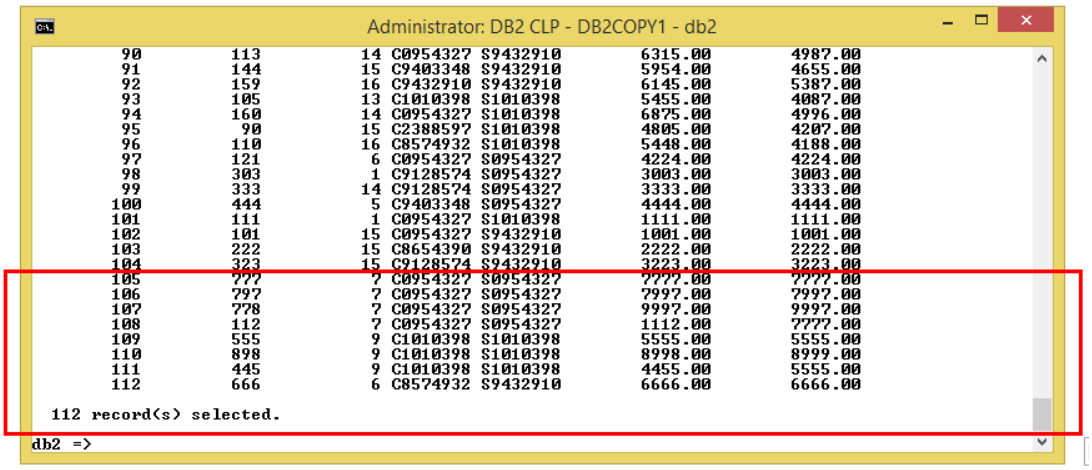
#	SalesUnits	CustID	StoreID	ItemID	Day	Month	Year	SalesDollar	SalesCost
1	777	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7777.0	7777.0
2	797	C0954327	S0954327	I0036566	3.0	7.0	2011.0	7997.0	7997.0
3	778	C0954327	S0954327	I0036566	3.0	7.0	2011.0	9997.0	9997.0
4	112	C0954327	S0954327	I0036566	3.0	7.0	2011.0	1112.0	7777.0
5	555	C1010398	S1010398	I0036566	1.0	2.0	2012.0	5555.0	5555.0
6	898	C1010398	S1010398	I0036566	1.0	2.0	2012.0	8998.0	8999.0
7	445	C1010398	S1010398	I0036566	1.0	2.0	2012.0	4455.0	5555.0
8	666	C8574932	S9432910	I0036577	1.0	5.0	2011.0	6666.0	6666.0

Figure 62: Final output of SSSales step partially shown

Step 8 – Go to DB2 Cloud so you can verify the number of rows in the *SSSales* table. Type the following given statement:

- SELECT * FROM SSSALES;

You should see 112 rows with 8 new rows added to the 104 rows of *SSSales* table (Figure 63).



Administrator: DB2 CLP - DB2COPY1 - db2

```
    90      113      14 C0954327 S9432910      6315.00      4987.00
    91      144      15 C9403348 S9432910      5954.00      4655.00
    92      159      16 C9432910 S9432910      6145.00      5387.00
    93      105      13 C1010398 S1010398      5455.00      4087.00
    94      160      14 C0954327 S1010398      6875.00      4996.00
    95       90      15 C2388597 S1010398      4805.00      4207.00
    96      110      16 C8574932 S1010398      5448.00      4188.00
    97      121       6 C0954327 S0954327      4224.00      4224.00
    98      303       1 C9128574 S0954327      3003.00      3003.00
    99      333      14 C9128574 S0954327      3333.00      3333.00
   100      444       5 C9403348 S0954327      4444.00      4444.00
   101      111       1 C0954327 S1010398      1111.00      1111.00
   102      101      15 C0954327 S9432910      1001.00      1001.00
   103      222      15 C8654390 S9432910      2222.00      2222.00
   104      323      15 C9128574 S9432910      3223.00      3223.00
   105      777       7 C0954327 S0954327      7777.00      7777.00
   106      797       7 C0954327 S0954327      7997.00      7997.00
   107      778       7 C0954327 S0954327      9997.00      9997.00
   108      112       7 C0954327 S0954327      1112.00      7777.00
   109      555       9 C1010398 S1010398      5555.00      5555.00
   110      898       9 C1010398 S1010398      8998.00      8999.00
   111      445       9 C1010398 S1010398      4455.00      5555.00
   112      666       6 C8574932 S9432910      6666.00      6666.00

  112 record(s) selected.

db2 =>
```

Figure 63: Rows added to DB2 SSSales table coming from Access table.